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SCHOOL OF ENGINEERING SCIENCES

FINAL YEAR PROJECT REPORT

ON

AN IOT-ENABLED SMART DOOR FOR MONITORING BODY TEMPERATURE AND FACE MASK DETECTION

BY

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THIS PROJECT REPORT IS SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE

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Abstract

The outbreak of the deadly COVID-19 virus has been a big worry to the world, and as a result there is the need to take precautions at every place. Because of the COVID-19 pandemic, wearing a mask is required in broad daylight spaces, as appropriately wearing a mask offers a greatest preventive impact against viral transmission. Body temperature has likewise become a crucial parameter in determining the health status of an individual. In public places where there are lots of crowd, it is very difficult to manually monitor and ensure every individual is abiding by the COVID 19 protocols. In this work, an IoT-enabled smart door that utilizes a machine learning model for checking body temperature and face mask detection. This proposed model can be utilized for any shopping center, lodging, airports, banks, apartment entrances and so forth. As a result, a cost-effective and reliable method of utilizing computer vision and sensors to build a healthy environment. The project comprises of an integration of hardware and software. The proposed system uses contactless infrared temperature sensors to measure the temperature of the individual at the entrance of the public places. Our system also implements real time deep learning models for face mask detection and classification. The module has three classification instances which includes; individuals who wear face mask properly, improperly and without a face mask using MobileNetV2, CNN and Tensor flow libraries using the transfer learning approach. Access is granted to the individual if the output from the temperature measurement module and the face mask detection module meet the acceptable threshold that is predetermined by the system. Also, the system is interfaced with a web application that keeps records of individuals for further control and monitoring.

ACKNOWLEDGEMENT

We first express our deepest gratitude to God Almighty who has given us grace and strength and without Him, our project will not be a success.

We also owe and express our deepest gratitude to our project supervisor, Dr. Robert A. Sowah who took keen interest in our project work and guided us all along, till the completion of our project work by providing all the necessary information for developing a fully functional system. We are also grateful for the emotional support he gave us at every point in time.

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Chapter 1 - Introduction

1.0 Background

The coronavirus disease also known as Covid 19, originated from Wuhan, China and has rapidly spread across several countries including Ghana. Living at home, working from home, self-driving, keeping away from affected cities are all possibilities for restricting the propagation of the virus. Individuals who have COVID-19 contamination can have a wide scope of indications, which range from gentle to extreme sickness. Indications of COVID-19 infection can show up from two to 14 days after openness to the infection. Fever, dry cough, sore throat, migraine, muscle strain, clog or runny nose, loose bowels, and drowsiness are the main normal symptoms of COVID-19. The significant ways for the infection's spread, as per WHO, are respiratory drops and actual contact between individuals. The respiratory droplets from one person who is in contact with the infected one can be transferred to other in different forms. As indicated by the World Health Organization(WHO), since December 2019, in excess of 114 nations experienced Covid 19 pandemic which has proclaimed as a dangerous illness that has intentionally contaminated more than 2.43 million deaths worldwide as on February 18, 2021. Figure 1 shows the COVID 19 spread model.

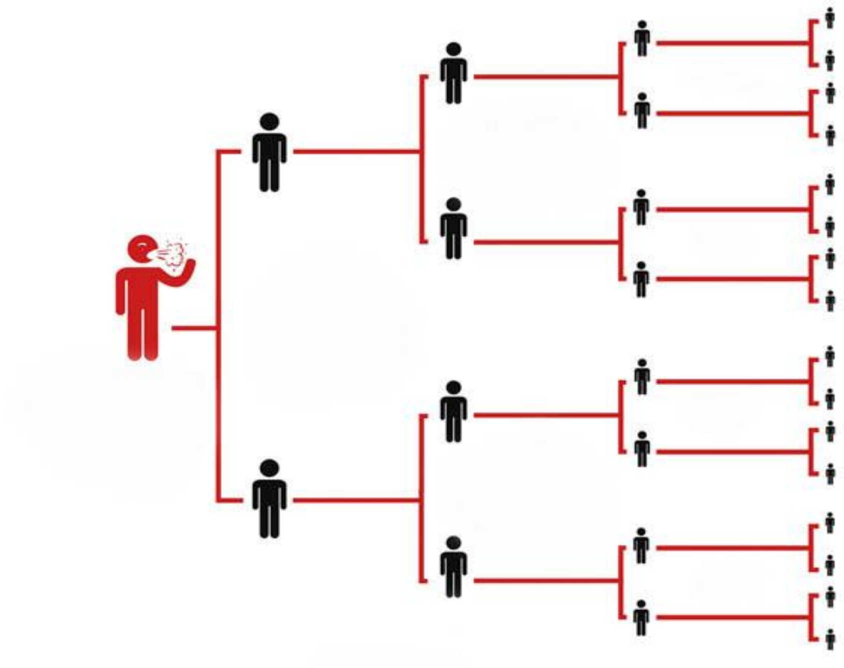

Figure 0-1: COVID 19 spread model

With an enormous populace and poor settlement in most towns, Ghana would experience difficulty forestalling the spread of the Covid.19. In our part of the world, in our fight against covid 19 little advancements have been made in the provisions of enough vaccines to overcome the virus. Government carried out an assortment of security and wellbeing drives to lessen disease transmission, including social distancing, mandatory indoor mask-wearing, quarantine, restricting citizens' traveling within state boundaries and abroad, self-isolation, and the avoidance and abrogation of huge social events and gatherings. Technology has additionally assumed a basic part in aiding battle the spread of the infection. The use of thermometer guns, automated hand sanitor dispensers, disinfection tunnels, social distancing and contact tracing applications have demonstrated to be vital in the means taken towards battling the infection. Schools have additionally been compelled to use contactless form of teaching and learning by using online learning platforms. However, more can still be done with technology to help minimize the spread of the virus and help rid the world of this pandemic. So, in our proposed technique, we've created a mask recognition algorithm that can distinguish a wide variety of mask kinds and designs. With the help of computer vision, the deep learning algorithm is applied and the TensorFlow, Keras library of python is used for implementation. The Contactless Sanitization and Temperature Monitoring System are outfitted with a contactless Infrared Temperature Sensor MLX90614, an OLED show, Audio speaker module, Active IR sensor, TIP32C PNP Transistor, DC Water siphon, LED Indicators and Raspberry pi to detect determine whether a person's temperature is normal or abnormal. While a few measures are being taken at the State and Central level to battle the circumstance, it has become the need of great importance, basically for the functioning populace to get out of the solace of their homes to support their living and just as to determine the affordable irregularity. With these reasons on the front, the proposed model will absolutely assist with guaranteeing the security and wellbeing health of the multitude of representatives when directed in their associations.

1.2 Problem Definition

The COVID 19 disease has brought a national crisis with its deadly spread of over 95,000 confirmed cases across the country, according to the Ghana Health Service.

a. Wearing of face masks is a main possible way of dealing with this pandemic.
b. However, it is observed that some people still do not wear facemasks when visiting public places due to some inexcusable reasons and they end up exposing the general public to the potential spread of the virus.
c. There is therefore a demand for automated intelligent systems that can provide effective surveillance at public places by ensuring people are always wearing their nose masks.

1.3 Project Objectives

1.3.1 General Objectives

a. Provide an automated way of ensuring the wearing of nose masks at public places

b. Check and monitoring the body temperature of individuals.

c. Build a web application to keep records of individuals for purpose of contact tracing

1.3.2 Specific Objectives

a. Build a facial detection system that is able to detect a face from a scene using transfer learning

b. Use of computer vision techniques to identify faces with or without nose masks

c. Measure the temperature of individuals using IR temperature sensors

d. After processing, send a signal to the door control mechanism to indicate whether the door should be opened or closed

e. Transfer data of individuals from the IoT system to the web application.

1.4 Relevance of Project

This project seeks to minimize the rate of spread of the corona virus by providing IoT based contactless automated temperature, face mask detection and enforcement systems at a minimal cost.

1.5 Outline of the Thesis

This report is divided into five (5) chapters. A description of each chapter is as shown below: Chapter 1 presents the introductory background of an IoT-enabled smart doors for monitoring body temperature and face mask detection, and how our proposed system aids in combatting this deadly virus. This chapter also focuses on the motivation behind our proposed system, the relevance of this project and the objectives of our proposed system.

Chapter 2 focuses on a review of existing systems to recognize the work previously done while evaluating their strengths and shortcomings and identifying existing gaps to make vital enhancements in this project.

Chapter 3 outlines the methodology of the design and development of the whole system.

Chapter 4 shows the ways in which the individual components were integrated and tested by various performance evaluation methods.

Chapter 5 presents the conclusions of the system development efforts. It captures the challenges faced in the various system design stages as well as key recommendations.

Chapter 2 – Literature review

2.0 Introduction

IoT is being utilized in different fields, similar to control and observing mechanical, electronic, keen security, and medical services frameworks. In this part, we depicted different IoT-based frameworks, in which the creators proposed various procedures for various fields of life to work on the expectations for everyday comforts. Starting in December 2019, the abrupt new sort of Covid pneumonia (COVID-19) immediately seethed across China and surprisingly the world [1]. As of July 15, 2020, more than 13.65 million affirmed cases have been accounted for in excess of 220 nations and districts all throughout the planet, and in excess of 580,000 patients have kicked the bucket. As of now, it is as yet proceeding to spread for a huge scope [2]. The new sort of Covid is exceptionally irresistible. It tends to be spread through contact, droplets, aerosols and different carriers noticeable all around, and it can get by for 5 days in a reasonable climate [3]- [4]. The "Rules for the Prevention of New Coronavirus Infection Pneumonia" given by the Public Health Commission accentuated that when people go out to public spots, look for clinical treatment and

take public transportation, they need to wear face masks or N95 masks to forestall the spread of the infection. Along these lines, it is everybody's obligation to wear masks out in the open spots during the pandemic, yet this requires the cognizant consistence of the individual, yet additionally the reception of specific measures to direct and oversee. As of now, in spite of the fact that there is no algorithm explicitly applied to face mask wearing detection, with the advancement of deep learning in the field of computer vision [5-7], neural network based objective identification calculations are utilized in common objective recognition, face detection, and remote detecting picture targets. Location, clinical picture, location and regular scene text identification are generally utilized in fields [8]- [11]. Face recognition calculations depend on a serious level of acknowledgment exactness, and have tremendous application potential in homeroom participation, character confirmation, access control frameworks, login and opening, and online media stages [12].

2.1 Review of existing works

- Temperature and Mask Scanning System by Uzma Takrim1, Farinnaj Sheikh, Nahid Sheikh, Masarrat Bano, Sanhita Hazra

Uzma Takrim et al [13] developed an IoT system that automates the process of Temperature Check-ups using Facial Land marking and Contactless IR Temperature Sensor and Mask Detection using Deep Learning Neural networks. Their system measures the human temperature of a person as well as scans persons to detect the wearing of nose masks or not. If it detects a person without nose mask approaching the camera sensor, the body temperature is displayed on an LCD and an alarm is sounded to warn and remind the person of not wearing a mask. Also if a temperature close to fever temperature is detected, the camera makes a quick response and informs the officer by an alarm message.

- Face Mask Detection Alert System using Raspberry Pi by Meghana Shinde1, Tanvi Sukhadare, Soham Vaidya, Prof. Meghali Kalyankar [14].

Meghana Shinde1, Tanvi Sukhadare, et al [14] developed a facial mask detection alert system using raspberry pi which distinguishes whether an individual has worn a face mask or not. Their system detects faces using a pi camera input and also detects if a person has worn face mask or not. If face mask policy is breached, an email alert with message and screenshot is sent to the officers in charge. Also, their system keeps records of screenshots of all “no mask” instances in the storage device of the raspberry pi.

- Design of an IoT Based Independent Pulse Oximetry kit as an early detection tool for Covid-19 Symptoms [15].

Development of pulse oximetry devices using IoT technology [4] as devices for real time monitoring of covid infected persons via smartphones with regards to them obeying the physical and social distancing protocols was used to keep track of the body temperature of the individuals.

- Deep learning implementation of face mask and physical distancing detection with alarm systems by Sammy V. Militante, Nanette V. Dionoso [16].

Implementation of face detection with alarm system for social distancing using deep learning techniques using CNN is discussed in [5]. Sammy V. Militante et all introduced a high accuracy algorithm to be used for detecting face masks fully based on CNN, binomial cross-entropy and gradient descent.

- A hybrid deep transfer learning model with machine learning methods for face mask detection in the era of the COVID-19 pandemic by Loey, M.; Manogaran, G.; Taha, M.H.N.; Khalifa, N.E.M.

The authors of [17] proposed a mixture model for face mask detection. Three datasets are utilized, which are the Simulated Masked Face Dataset (SMFD), the Real-World Masked. Face Dataset (RMFD), and the Labeled Faces in the Wild (LFW). Their proposed model comprises of two sections. The initial segment depends on ResNet-50 for the feature extraction from the picture, while the subsequent part is for the arrangement of either a face mask or without a face mask. The three algorithms are utilized for the characterization, which is Support Vector Machine (SVM), decision tree, and ensemble algorithm. The SVM algorithm accomplished the most noteworthy precision of close to 100% exactness on the LFW dataset.

- IoT-Based system for covid-19 indoor safety monitoring by Nenad Petrovic and Dorde Kocic

Nenad Petravoic and Dorde Kocic [18] proposed a system of which their solution consisted of a temperature measurement subsystem based on Adruino Uno, computer vison subsystem for mask detection and social distancing check based on Raspberry pi 3 and smartphones application for security guards. They relied on Adruino Uno equipped with infrared thermometer to monitor the temperature of Indi duals entering the building. Ardruino sends a signal to open the door if the temperature of the individual does not exceed the threshold value of the temperature set by the system. Their system also informs the security guards about the location of people not wearing nose mask through the mobile application.

- An Automated System to Limit COVID-19 Using Facial Mask Detection in Smart City Network by Mohammad Marufur Rahman, MD. Motaleb Hossen Manik , Md Milon Islam , Saifuddin Mahmud

The authors of [19] proposed an automated smart system for checking persons who are not wearing nose mask. CCTV cameras are used to capture images from public spaces, and these images are input into the system that detects if a person inside the image is without a facemask. If true, information about the person is sent to the proper authority for necessary actions to be taken.

- 2.1.8 An IoT-Based System for Automated Health Monitoring and Surveillance in Post-Pandemic Life [20]

Amir Fotovwat, Mohammed Reza Mohebbian et al [9] developed a COVID-SAFE platform relies that is made up of three segments which include a wearable IoT device, smartphone app and a cloud server. The software phase includes an Application Program Interface (API) which provides a platform for interacting with the users on a smartphone and a fuzzy decision making system installed on the cloud server. The hardware phase is made up of nodes developed on a Raspberry Pi Zero which collect specific vital data from the users and send to the decision making system. Advices such as the need to see a doctor, to maintain physical distance from others and avoidance of high risk areas are then sent to the user through the API.

Below is a table that identifies the strengths and weaknesses of the various works we used in developing our system

Table 1: Review of Existing Systems

Existing System	Title and Authors of the papers	The methodology employed	Strength(s)	Weakness(es)
Existing system [13]	Temperature and Mask Scanning System by Uzma Takrim, Farinnaj Sheikh, Nahid Sheikh, Masarrat Bano, Sanhita Hazra	Their system measures the human temperature of a person as well as scans persons to detect the wearing of nose masks or not.	The system makes use of a contactless temperature scanner and a mask monitor to determine if a person is granted access or not.	The system cannot determine if a person is correctly wearing a nose mask or not.
Existing system[14]	Face Mask Detection Alert System using Raspberry Pi by Meghana Shinde, Tanvi Sukhadare, Soham Vaidya, Prof. Meghali Kalyankar .	Their system detects faces using a pi camera input and also detects if a person has worn face mask or not. If face mask policy is breached, an email alert with message and screenshot is sent to the officers in charge. Also, their system keeps records of screenshots of all “no mask” instances in the storage device of the	Their system used MobileNetV2 and the Single Shot Detector(SSD) model to train the dataset and it gives better accuracy for simple masked face recognition	Their system is not able to detect people with high temperatures.

		raspberry pi.		
Existing System [15]	Design of an IoT Based Independent Pulse Oximetry kit as an early detection tool for Covid-19 Symptoms .	Development of pulse oximetry devices using IoT technology [4] as devices for real time monitoring of covid infected persons via smartphones with regards to them obeying the physical and social distancing protocols was used to keep track of the body.	Their proposed system can be used to monitor covid-19 patients through smartphones/PC with regards to physical and social distancing protocols.	The system does not yet have a system that can be analyzed in real time and keep records of the measurements from the patients.
Existing system [16]	Deep learning implementation of face mask and physical distancing detection with alarm systems by Sammy V. Militante, Nanette V. Dionoso [4].	Implementation of face detection with alarm system for social distancing using deep learning techniques using CNN is discussed in [5]. Sammy V. Militante et all introduced a high accuracy algorithm to be used for detecting face masks fully based on CNN, binomial cross-entropy and gradient descent	Their system used deep learning technique like Keras and CNN which obtain an azzuracy of 97.87% when detecting a paeson wearing face mask and 98.75% when detecting person without face mask	Their system lacked a temperature detection system and as a result it cannot check whether a person has high fever or not which is a mild symptom of the disease
Existing	A hybrid deep	Their proposed model	Their proposed	The system does

System [17]	transfer learning model with machine learning methods for face mask detection in the era of the COVID-19 pandemic by Loey, M.; Manogaran, G.; Taha, M.H.N.; Khalifa, N.E.M.	comprises of two sections. The initial segment depends on ResNet-50 for the feature extraction from the picture, while the subsequent part is for the arrangement of either a face mask or without a face mask. The three algorithms are utilized for the characterization, which is Support Vector Machine (SVM), decision tree, and ensemble algorithm. The SVM algorithm accomplished the most noteworthy precision of close to 100% exactness on the LFW dataset.	system archived 100% testing accuracy and was able to identify faces with or without nose mask.	not yet have a system that can be analyzed in real time and keep records of the measurements from the patients.
Existing system [18]	IoT-Based system for covid-19 indoor safety monitoring by Nenad Petrovic and Dorde Kocic	They relied on Adruino Uno equipped with infrared thermometer to monitor the temperature of Indi duals entering the building. Ardruino sends a signal to open the door if the temperature of the individual does not	Their system was able to do a social distancing check and it is interfaced with an application which keeps records of temperatures measured and alert the security guards	Their system lacked a temperature detection system and as a result it cannot check whether a person has high fever or not which is a

		exceed the threshold value of the temperature set by the system. Their system also informs the security guards about the location of people not wearing nose mask through the mobile application.	of the location of individuals who do not meet the system's requirements	mild symptom of the disease
Existing system [19]	An Automated System to Limit COVID-19 Using Facial Mask Detection in Smart City Network by Mohammad Marufur Rahman, MD. Motaleb Hossen Manik, Md Milon Islam, Saifuddin Mahmud	The authors of [8] proposed an automated smart system for checking persons who are not wearing nose mask. CCTV cameras are used to capture images from public spaces, and these images are input into the system that detects if a person inside the image is without a facemask. If true, information about the person is sent to the proper authority for necessary actions to be taken.	Their system archieved a high accuracy of 99.4%	The system does not yet have a system that can be analyzed in real time and keep records of the measurements from the patients.
Existing System[20]	An IoT-Based System for	Amir Fotovwat, Mohammed Reza	he IoT node tracks health parameters,	Their system lacked a

	Automated Health Monitoring and Surveillance in Post-Pandemic Life	Mohebbian et al [9] developed a COVID-SAFE platform relies that is made up of three segments which include a wearable IoT device, smartphone app and a cloud server. The software phase includes an Application Program Interface (API) which provides a platform for interacting with the users on a smartphone and a fuzzy decision making system installed on the cloud server. The hardware phase is made up of nodes developed on a Raspberry Pi Zero which collect specific vital data from the users and send to the decision making system. Advices such as the need to see a doctor, to maintain physical distance from others and avoidance of high risk areas are then sent to the user	including body temperature, cough rate, respiratory rate, and blood oxygen saturation, then updates the smartphone app to display the user health conditions. The app notifies the user to maintain a physical distance of 2 m (or 6 ft), which is a key factor in controlling virus spread.	temperature detection system and as a result it cannot check whether a person has high fever or not which is a mild symptom of the disease

Existing system[21]	IoT- Enabled smart doors for monitoring Body temperature and face mask detection by B Varshini, HR Yogesh, Syed Danish Pasha, Maaz Suhail, V Mashumitha, Archana Sasi	Their system measures the human temperature of a person as well as scans persons to detect the wearing of nose masks or not.	The system makes use of a contactless temperature scanner and a mask monitor to determine if a person is granted access or not.	The system does not yet have a system that can be analyzed in real time and keep records of the measurements from the patients.

2.2 The proposed work

This project aims to develop an IoT- enabled smart doors for monitoring body temperature and face mask detection system is to contain the spread of Covid-19 in public places such as shopping malls, offices, airports, banks and so on. The system can check an individual's body temperature and can perform face mask detection. The system is also interfaced with a web application that keeps records of individuals for further control and monitoring.

2.3 Scope of the project work

The study in this paper focuses on developing a real time IoT enabled smart door for temperature and face mask detection with an additional desktop application for monitoring real time data from the system.

Chapter 3 - System design and development

3.0 Introduction

A general approach to deal with our design is the integration of the face mask detection module and temperature detection module on to a raspberry pi which automates the opening of the door and also a web app which keep records of an individual for further control and monitoring.

3.1 System overview and functions

The system primarily comprises of 2 main subsystems, which are:

i. The hardware subsystem
ii. The software subsystem

3.1.1 The hardware subsystem

The hardware subsystem comprises of a raspberry pi, raspberry pi camera, an IR sensor, a temperature sensor and servo motor. The IR sensor is used to detect proximity of people to the device and the temperature sensor is used to measure the temperature of the person. These measurements are recorded and sent to a local server.

3.1.2 The software subsystem

The software subsystem consists of two subsystems. The first subsystem is a face mask detector application developed using CNN, MobileNetV2, AI and Machine learning technologies. This application is interfaced with a camera to detect if a person walking in is wearing a mask, not wearing a mask, or wearing it wrongly. This information is also stored in the local server.

The second subsystem is a desktop application that was developed using Pycharm. The front end was developed using the PyQt6 framework and the back end was developed using Django. The application is meant to be used by only administrators. The administrator can view the current global covid 19 statistics, the temperature and mask status of all persons who use the facility and also make decisions on which people to isolate and quarantine.

The software subsystem consists of a web application that is developed using Visual Studio code. The front end was developed using the react JS framework and the back end is developed based on the express JS framework. Users of the web application can view the location of the shuttle via Google maps and pay their shuttle fares using either mobile money or credit card. The credentials of users that is used for authorization are stored in a firebase database.

3.2 Requirements, analysis and specifications

3.2.1 System Requirements

i. The system to be developed must be able to:
ii. Detect faces with mask, without mask and improper wearing of face masks using raspberry pi camera.
iii. Transfer the pictures taken with the raspberry pi camera to a cloud server.
iv. Monitor the body temperature of an individual entering the premises.
v. Automate the opening of doors after detecting faces with mask and monitoring the body temperature.
vi. Display all the collected data from the cloud on the dashboard in the desktop application.

3.2.2 Functional Requirements

vii. Temperature measurement

The system should be able to measure the temperature of the individuals within the specified range of distance from the device.

viii. Face mask detection

The system should be able to detect faces from the video frame. Subsequently, it should be able to detect faces with proper mask, improper wearing of mask and faces without face mask.

ix. Smart door automation

The system should be able to determine whether to open the door or not based on temperature and mask status of the individual.

x. Send real time measurement data to the cloud server.

After temperature and mask status are recorded, they should be sent to the cloud server.

xi. User authentication

The system should verify the identity of the administrator when he or she logs in the desktop application. This is done by the MySQL authentication service since the credentials of our users is stored in a MySQL database.

xii. Display the information from the cloud server on the dashboard of the desktop application.

The desktop application should retrieve data from the cloud server and display on the dashboard.

3.2.3 Non-functional requirements

xiii. Performance

The system has a short response time which means the hardware tracker is able to send the coordinates quickly and the web application is also able to receive the information from the database server quickly due to the efficient and optimal nature of the queries for these processes.

xiv. Reliability

The system is available to be used by the user all the time due to how the software and hardware subsystems are structured. This will make it the system to be user-friendly and be able to be used anytime during the day.

xv. Scalability

The system adapts to the changing needs and demands of the users. Due to the versatility of our system and the fact that more people will be using our system and also be requesting more services, adequate resources have been provided so that users can use our system.

xvi. Security

The system uses encryption to secure user data. Since our user data and system data are stored in a database, SQLite has security rules that protect data from being edited or changed.

3.3 System design and development process

3.3.1 System Architecture

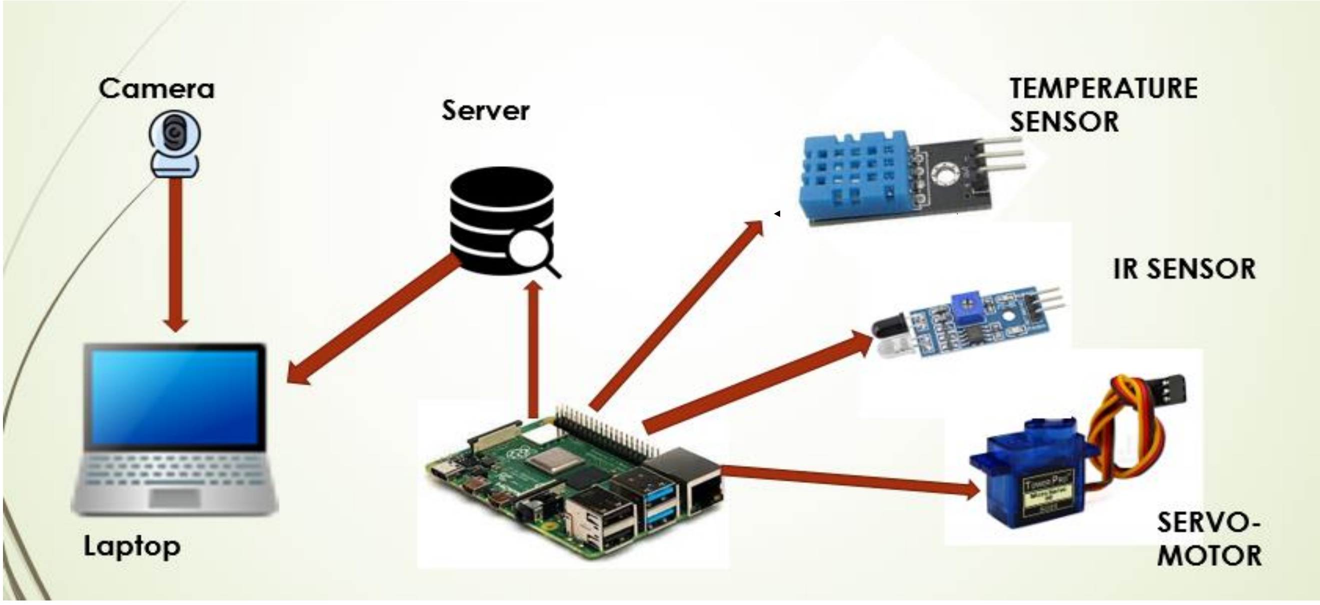

Figure 3.3-1 System Architecture

From the schematic above, the desktop application and the hardware temperature measurement system are connected using a server socket connection so they can communicate with each other. The Raspberry Pi 3b microprocessor controls the activities of the various sensor modules. The microprocessor obtains the temperature reading of a person when the hand is placed a few centimeters away from the infrared sensor. This data is then sent to the server through server socket connection. The server then feeds the desktop application with this data and it is displayed on the dashboard.

Flow diagram for hardware subsystem

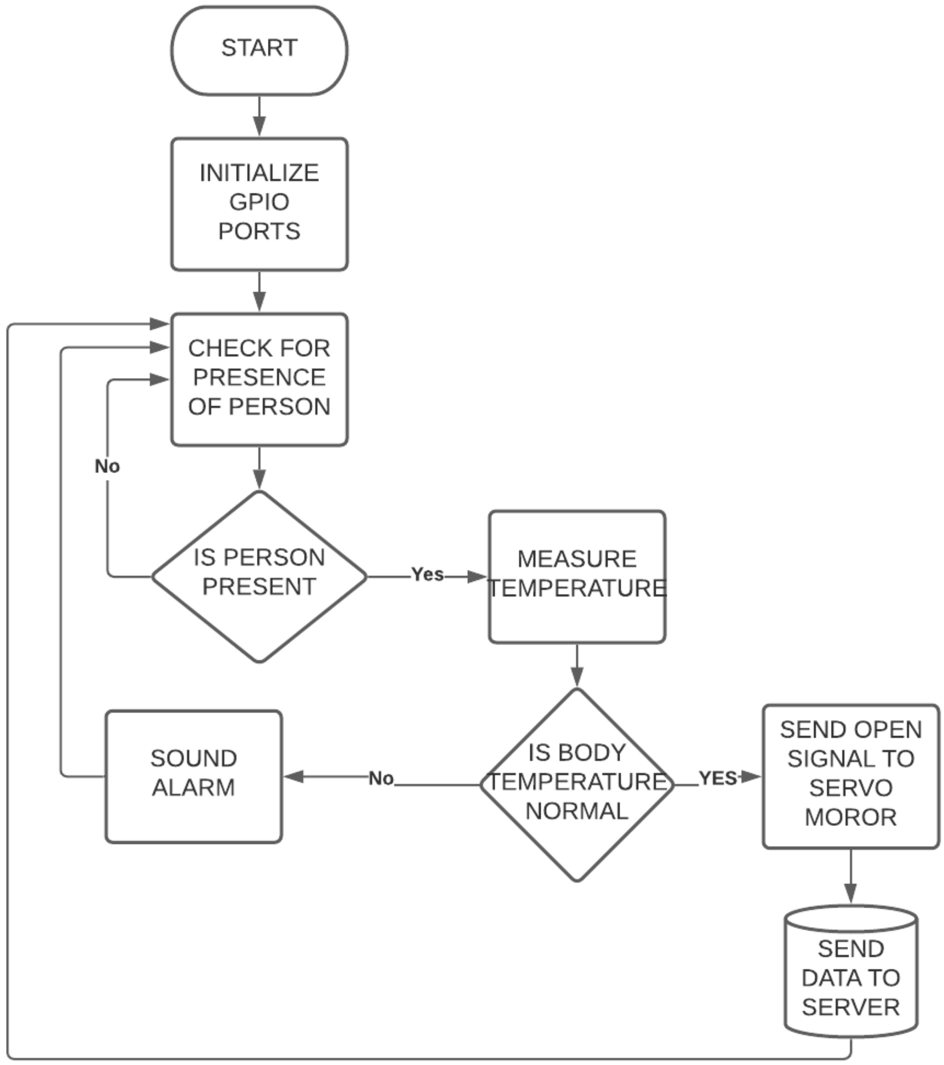

Figure 3.3-2 Flow diagram of the hardware subsystem

From the diagram above, when the hardware program starts running, all GPIO ports are initialized. The IR sensor continuously checks for the presence of a human being. If detected, the temperature of the person is measured. If the temperature reading is higher than the normal body temperature which is between 30 to 37 degree Celsius, an alert is sounded by the system and person is denied access to the building. If the reading however is within the healthy range, the door is opened through the servo motor and the temperature reading is sent to the server to be displayed in the desktop application.

3.3.2 Software subsystem

The software subsystem was developed in two modules, namely the machine learning module for detection of face masks and the desktop application module for overall operation of the system.

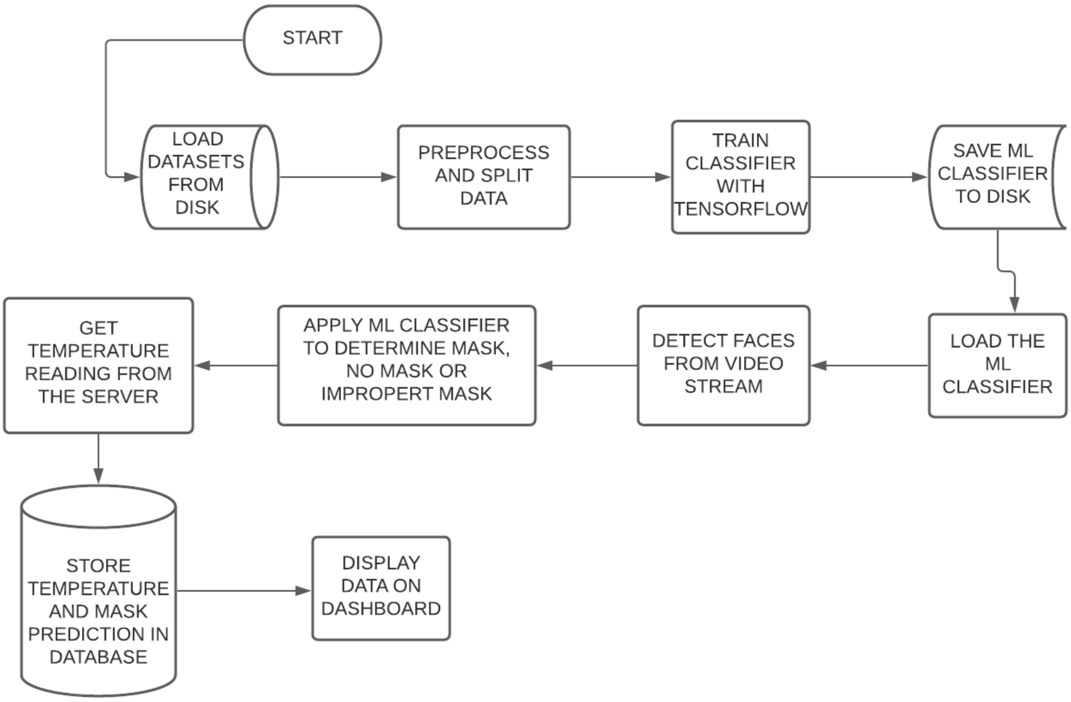

Figure 3.3-3: Flow diagram of the software subsystem

3.3.3 Methodology

Loading of Face Mask Datasets

The dataset, used for this project was accessed on the the Kaggle data repository. It has three different classes which include people with nose masks, people with no nose masks and people who wear masks improperly. In total the dataset consists of 2000 images of people wearing masks, 2000 images of people not wearing masks and 2500 images of people not wearing masks properly. The total number of images for this dataset sums up to 6,500 images. Figures 3.1,3.2 and 3.3 show samples of people wearing masks, people not wearing masks and people wearing masks improperly. In our dataset, people who had masks on were labelled with a class 0, people with no masks were labelled with class 1 and people with improper nose masks were labelled with class 2. This dataset was used training our classification model using transfer learning. Figure 7 shows the distribution of the total number of different categories of the dataset.

Figure 3.3-4: Sample of with mask dataset

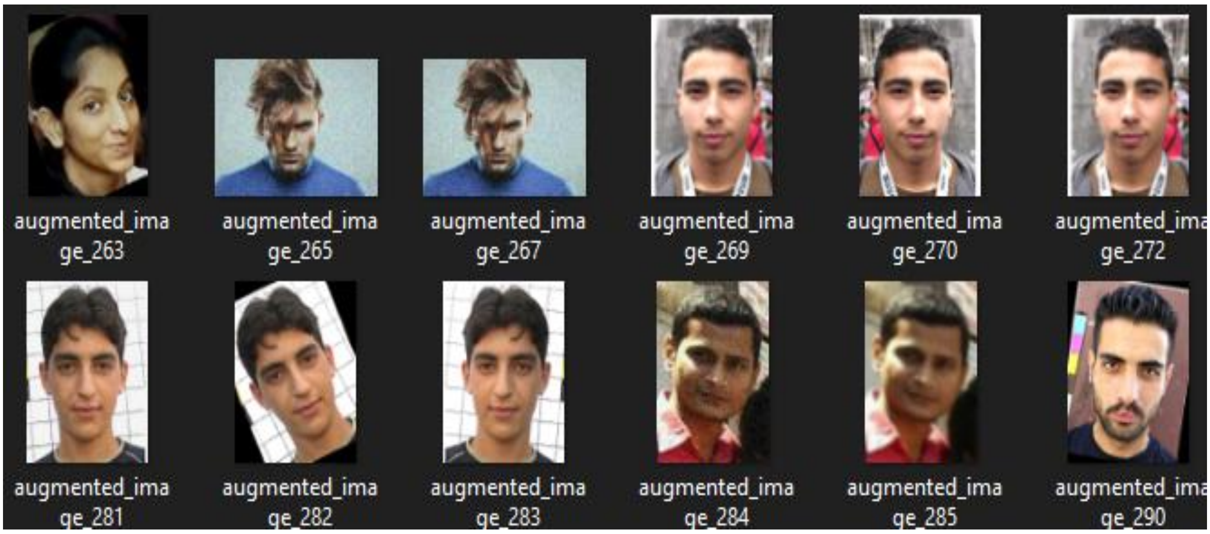

Figure 3.3-5: Sample of faces without mask dataset.

Figure 3.3-6: Sample of faces with improper mask dataset.

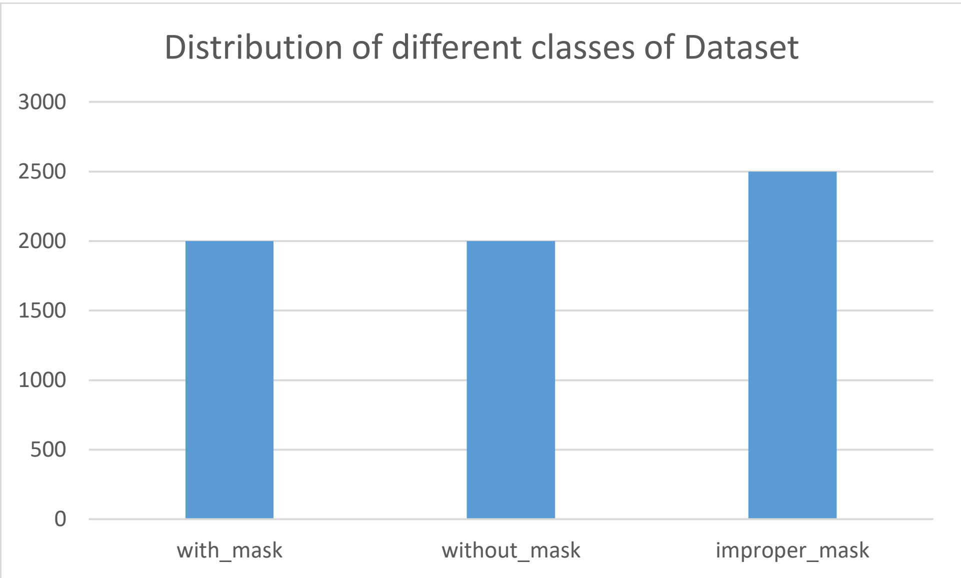

Figure 3.3-7: Distribution of different classes of Dataset

- Data pre-processing

Performing data pre-processing is very critical and necessary in machine learning as it improving the quality of data which helps the extraction of important features from the data. It involves getting the data cleaned and organised so they can be used for building and training a model. For the entire dataset, all images were converted to grayscale. This was necessary so as to simplify the algorithm as well as reduce the computational power involved in extracting the desired features from the images.

- Training

We utilized pre-trained deep learning models, which are MobileNetV2, and Convolutional

Neural Network (CNN), to detect a face with proper mask, face with an improper mask, and face without mask. We set the output layer of these models as non-trainable by freezing the weights and other trainable parameters in each layer to not be trained or updated when we feed our dataset. Further, we added an output layer to train on our dataset. This output layer would be the main trainable layer in our new model. We utilized the Adam optimizer with a learning pace of 0.001, cross-entropy for our misfortune, and exactness for our grid. After building the model, we label three probabilities for our results. [‘0’ as ‘with mask’,’1’ as without mask, and ‘2’ as ‘improper mask’]

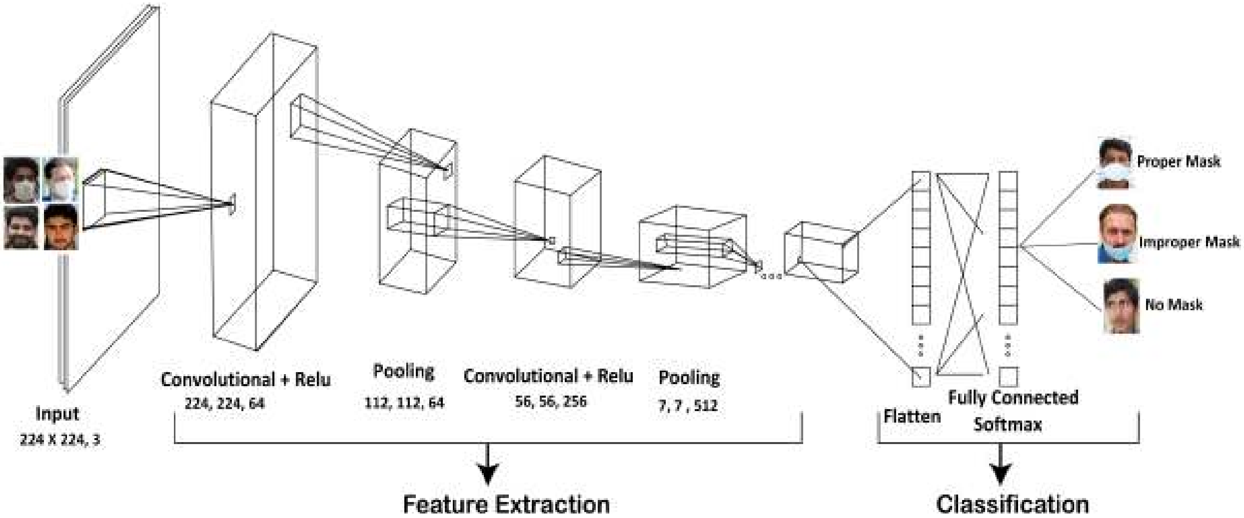

Figure 3.3-4 Architecture of the Convolutional Neural Network

The architecture of the CNN is made up of an input layer, an output layer, and multiple hidden layers. The hidden layers characteristically consist of convolution layers, pooling layers, fully connected layers and normalization layers (Relu). For this network, the input are the images of people with mask, without mask and with improper mask with a sizes of 224 by 224 and three RGB channels.

We have our first layer of the CNN as a convolutional layer with the Relu function defined below with a kernel size of 224 x 224

$$Relu(x)\ =\ max(0,x)$$

The next layer of the architecture is max pooling which includes contracting the image stack. To pool a picture, the window size is characterized as (112 x 112, 64 output channels). This window is then sifted across the strides’ structure, with the value being recorded for each window.

We have a flattened layer with a fully connected layer with the softmax function shown below

$$pi\ =\ e\ x\ i\sum j\ e\ x\ j$$

where xi is the total input received by unit i and pi is the prediction probability of the image belonging to class i. In this layer, all values give a prediction of which class the input image belongs to.

- SERIALIZE THE MODEL

After training our model, we serialized it and saved it to our local disk to be used by the desktop application.

3.3.4 Application Use Case Diagram

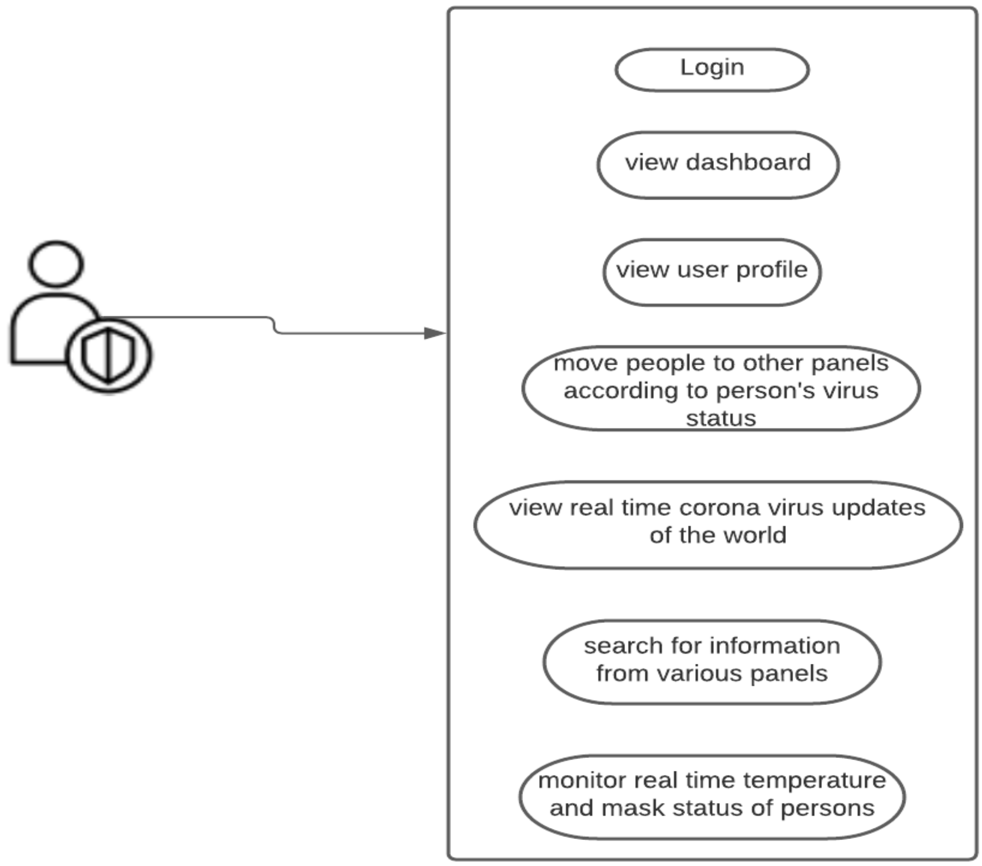

Figure 3.3-8: Application use case diagram

From the use case diagram above, the user can perform the following activities with the application;

i. Sign in as an administrator to the desktop application to start the system.

ii. View dashboard which shows various panels of the application

iii. View user profile

iv. Change person's status from suspected to quarantined if temperature is very high

v. Search for information from the database. Data like temperature, mask status, time of arrival and picture of a person from the database.

vi. View real time corona virus updates of the world.

3.3.4 Activity Diagram

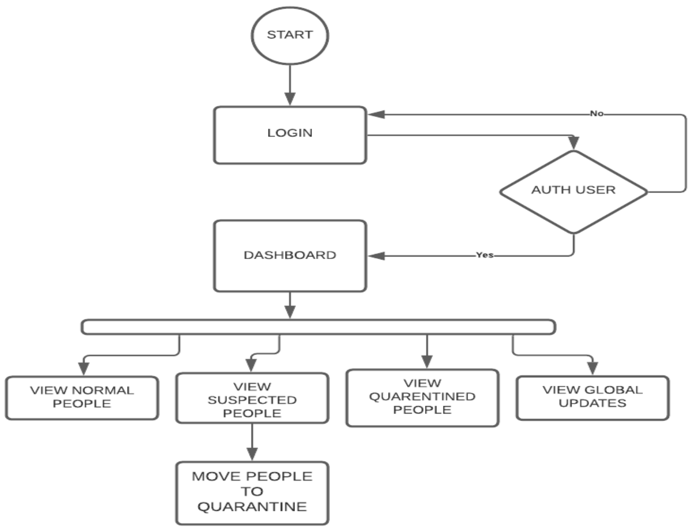

Figure 3.3-9: Activity Diagram

At first run, a default admin would be created and would be able to add other users. When the application is started, the admin is required to enter his credentials to access the system. The application authenticates the admin's credentials from the SQLite database. If authentication is not successful, the application denies the user access to the system and redirects him back to the login page. If the authentication is successful, the dashboard of the application is displayed. On the dashboard page, the admin can view his profile, add other users to the system, remove users and also update his credentials. Since this application is meant to be used by industries and in homes, it is not meant to be used by the general public. This means that only administrators can add users to the system. Also the user can view panel information such as the normal people, suspected people and the quarantined people based on their temperature measurement. The user can search for information by time, temperature, mask status and the picture of the persons who use the facility. In addition, users can view and monitor the real time global updates of the coronavirus.

3.3.5 Database Design

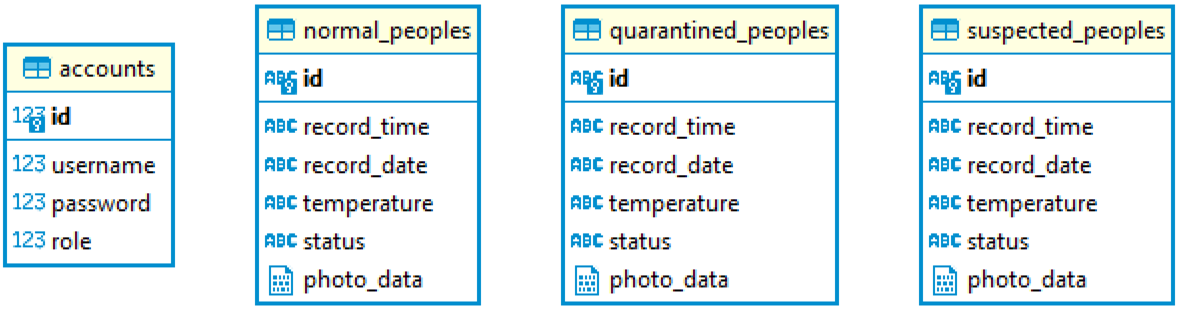

Figure 3.3-10: Database Design

A SQLite database was chosen for this project since it is fast and perfect for real time applications like our system.

3.4 System modelling

3.4.1 Schematic diagram

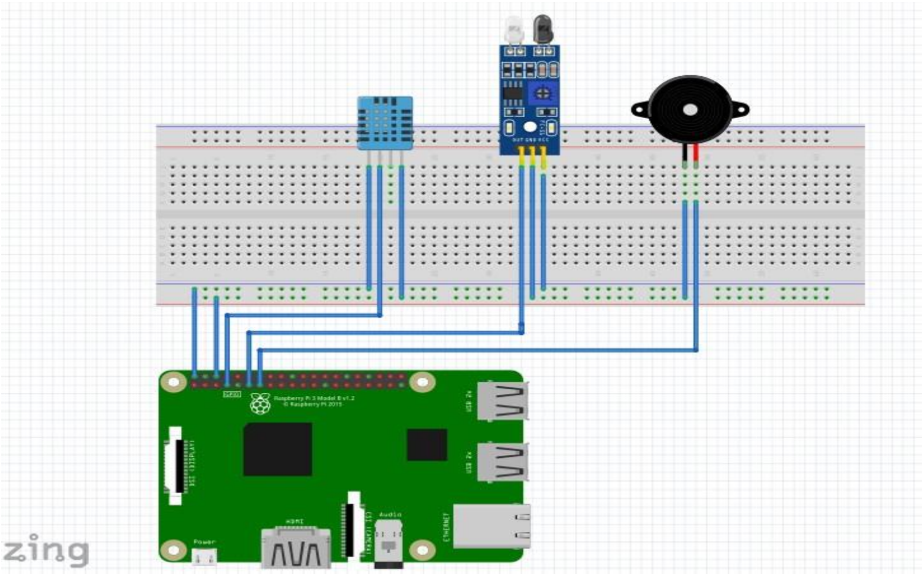

Figure 3.4-1: Schematic diagram of the hardware system

The diagram above was modelled using the fritzing application. From the diagram, a DHT11 temperature sensor, an Infrared(IR) sensor and a Piezo buzzer are connected to the Raspberry Pi 3b through the GPIO ports using jumper wires. The Raspberry Pi is connected to a 5V power source to enable it performs its functions.

3.5 Development tools and material requirements

3.5.1 Material Requirements

i. Raspberry Pi 3b

Figure 3.5-1: Raspberry Pi 3b

The Raspberry Pi is a minimal expense little PC that associates with a PC screen or TV and works with a customary keyboard and mouse. It is a helpful little contraption that spotlights on showing individuals of any age prearranging dialects like Scratch and Python. It can play out every one of the elements of a PC, for example, web surfing and survey more noteworthy definition clasp, worksheets, and messing around. It has been utilized in a few computerized gadgets, including tweeting perching spaces, music machines, and finders, just as climate stations and infrared cameras since it is equipped for communicating with the external climate. Profoundly and a video center mixed media co-processor and the GPU, which incorporates double center sight and sound co-processor, including a Bluetooth 4,1 (Bluetooth and Bluetooth Classic). With Bluetooth Low Energy (BLE) and BCM43143 Wi-Fi, the Raspberry Pi 3 offers an up-grade towards another primary processor and improved systems administration. Moreover, the force the board of Raspberry 3 has been improved, with a redesigned power supply with 3.5 Amps that can deal with all the more remarkable outside USB gadgets. The inherent USB ports of Raspberry Pi 3 give adequate network to interface the mouse or whatever else to the Raspberry Pi. Most Raspberry Pi framework chips can be overclocked to 800 MHz, and some can be overclocked to 1000 MHz It is accounted for that the Raspberry Pi 2 can be comparatively overclocked, in any event, arriving at 1500 MHz in outrageous cases (without all security highlights and overvoltage limitations). On Linux disseminations, you can utilize the program order to run "sudo raspi-config" to perform boot overclocking without breaking the guarantee. In specific examples, the Pi will consequently deactivate overclocking when the chip temperature arrives at 85°C to drop the robotized settings on overclock and overclocking (that will drop guarantee). A cooler can be utilized to forestall overheating of Raspberry pi.

3.5.2 IR Sensor

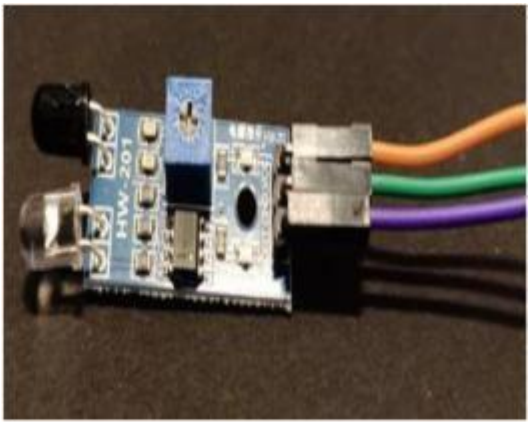

Figure 3.5-2 IR Sensor

Infrared sensors are utilized to count and screen the quantity of individuals who go into and leave the room. The IR sensor's working voltage is 5VDC, and the I/O pins are 5V and 3.3V viable. It accompanies an assortment of alternatives. Fig 3 depicts an IR Sensor that includes a constructed encompassing light sensor and a mounting opening, just as a flexible detecting scope of up to 20cm.

3.5.3 DHT11 SENSOR

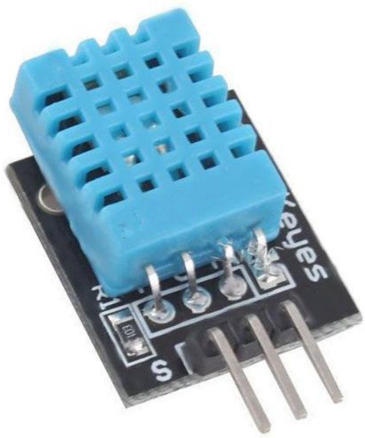

Figure 3.5-3 DHT11 SENSOR:

The DHT11 is a basic, ultra low-cost digital temperature and humidity sensor. It uses a capacitive humidity sensor and a thermistor to measure the surrounding air, and spits out a digital signal on the data pin (no analog input pins needed). Its fairly simple to use, but requires careful timing to grab data.

3.5.4 Other Components Used

a. Servo Motor

Figure 3.5-4a Servo Motor

A servo motor is used to demonstrate the opening and closing of the main door. shows a diagram of a Servo Motor, that produces velocity and torque based on the voltage and the amount of current supplied. It also works as a part of a closed-loop system providing velocity and torque as commanded from the servo controller with a feedback device to close the loop.

b. Jumper Wires

Figure 3.5-4b Jumper Wires

The jumper wires are used to connect the DHT11 temperature sensor, the IR sensor and the buzzer to the GPIO pins of the raspberry pi

c. Piezo Buzzer

A piezo buzzer is a type of electronic device that's used to produce a tone, alarm or sound. It's lightweight with a simple construction, and it's typically a low-cost product. Yet at the same time, depending on the piezo ceramic buzzer specifications, it's also reliable and can be constructed in a wide range of sizes that work across varying frequencies to produce different sound outputs.

Figure 3.5-4 :Piezo Buzzer

3.5.5 Development tools

Thonny IDE

The Thonny IDE is used to develop the code which is used on the Raspberry Pi microprocessor to control the processes of the temperature sensor, the IR sensor and the buzzer modules. This code is a written in python and runs directly on the microprocessor.

PyCharm IDE

The desktop application was developed using Visual Studio Code in the Python programing language. The front end of the desktop application was developed using the PyQt5 framework. The backend was developed using SQLite. The Ninja Coronavirus API is used to provide the real time covid data across the world.

Postman Software

The Postman Software is used to test API endpoints to see if they are working. It was used to test the Ninja Coronavirus API.

Chapter 4 – Implementation and testing

4.0 Introduction

This section shows how our system was developed, depicting how our subsystems were implemented and tested for their various functionalities.

4.1 System implementation process

4.1.1 The general system implementation process

i. The camera, together with the hardware module is placed at the entrance of the facility.
ii. The admin logs in and starts the desktop application.
iii. The camera starts to capture video frames and detects if there is a person in the frame.
iv. If a person is detected a photo of the person is captured and processed to see if he/she is wearing a mask or not.
v. If the person has on a mask measure the temperature of the person by prompting the person to take his hand close to the temperature sensor.
vi. The temperature of the person is then measured and sent to the database through python sockets.
vii. Person is allowed access to the building if temperature is within the healthy range of not more than 37 degrees Celsius.
viii. Display time and date of arrival, temperature and the status of the temperature.

4.1.2 Hardware subsystem implementation process

4.1.2.1 Then the various components were bought and connected, as shown in the system modelling section

4.1.2.2 The various components were then tested individually to check if they worked adequately using different python codes

4.1.2.3 Finally, the various python code for the hardware subsystem was written and run on the Raspberry Pi 3b microprocessor.

4.1.2.4 A 5V type B charger was used to power the microprocessor

4.1.3 Software subsystem implementation process

4.1.2.5 The software subsystem was developed using the PyCharm software in Python. It comprises of the desktop application and the database.

4.1.2.6 The front end of our application was developed using the PyQt5 framework, which makes the front end very rich and user friendly irrespective of the operating system running on that device.

4.1.2.7 Python sockets was used to receive temperature readings from the raspberry pi to the desktop application.

4.1.2.8 The Ninja coronavirus API was used to show the real time corona virus updates across the world.

4.2 Testing and results

4.2.1 Hardware Subsystem

4.2.1.1Temperature Device up and Running

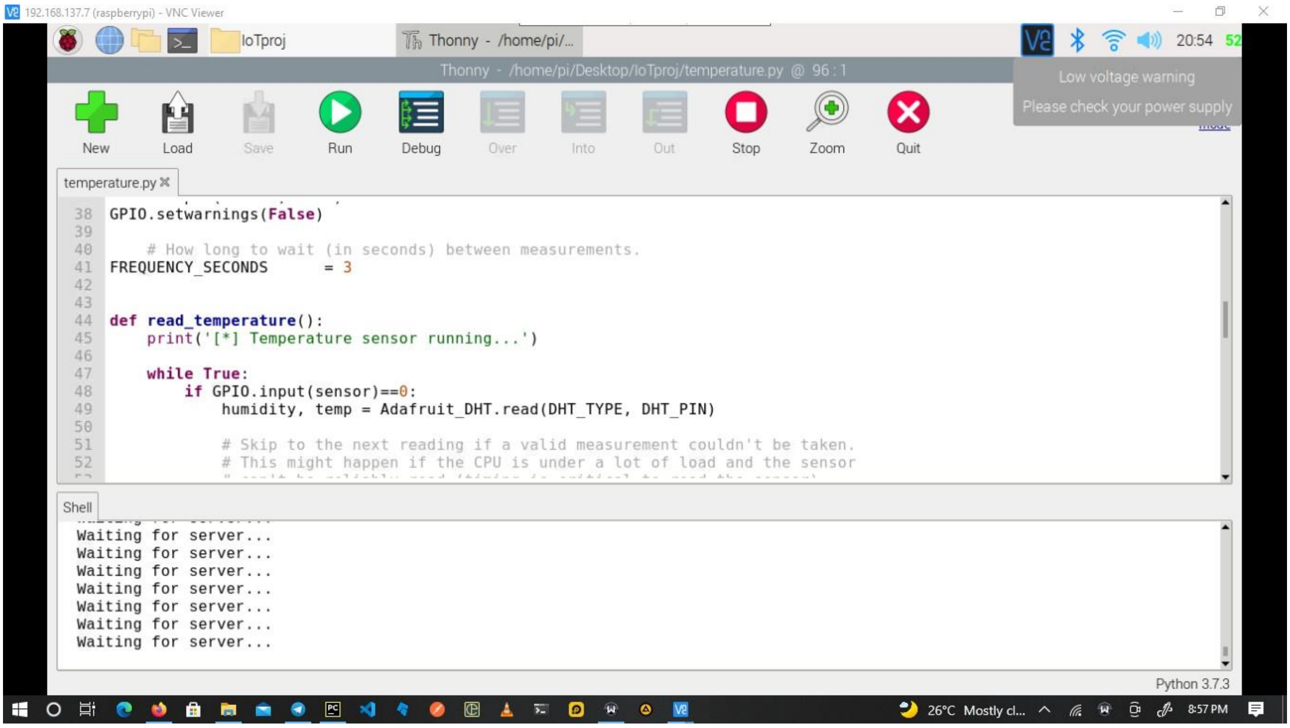

Figure 4-2-0-1: Waiting for desktop application to request temperature scan

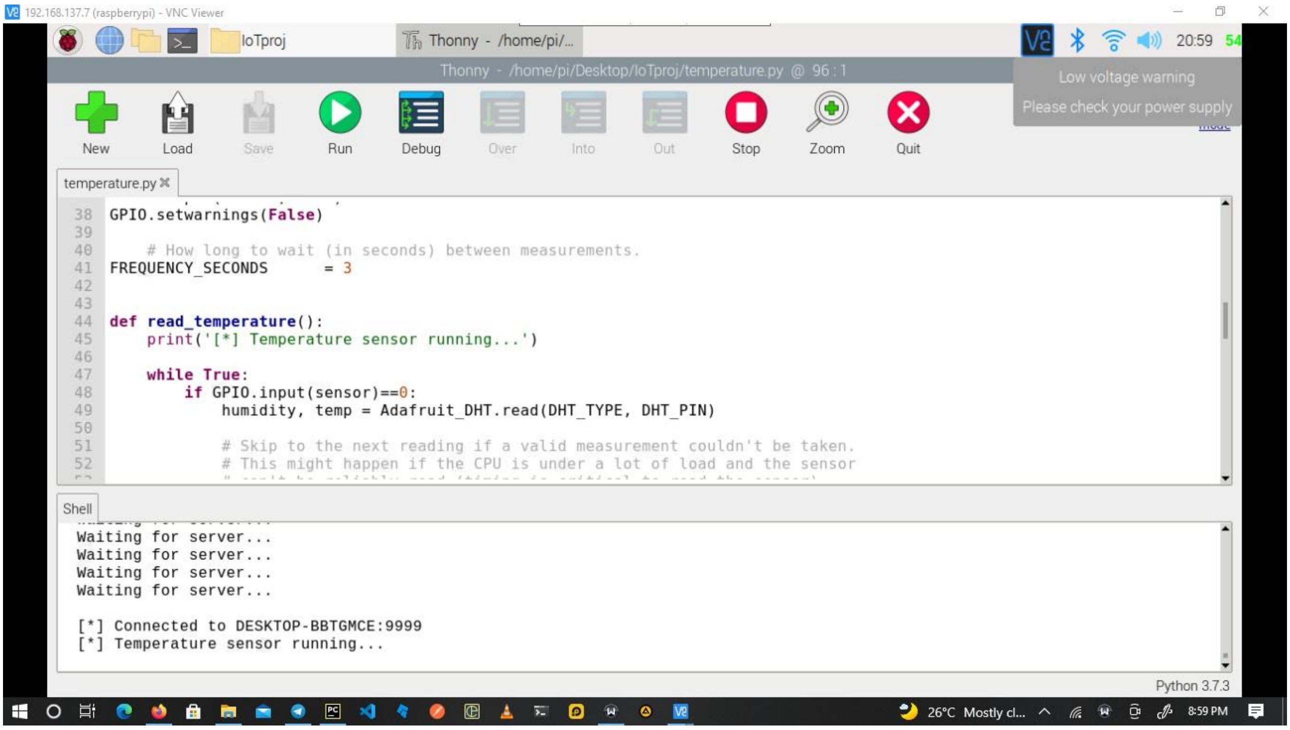

Figure 4-2-0-2: Temperature Device about to take a scan

Testing the Final Hardware Subsystem

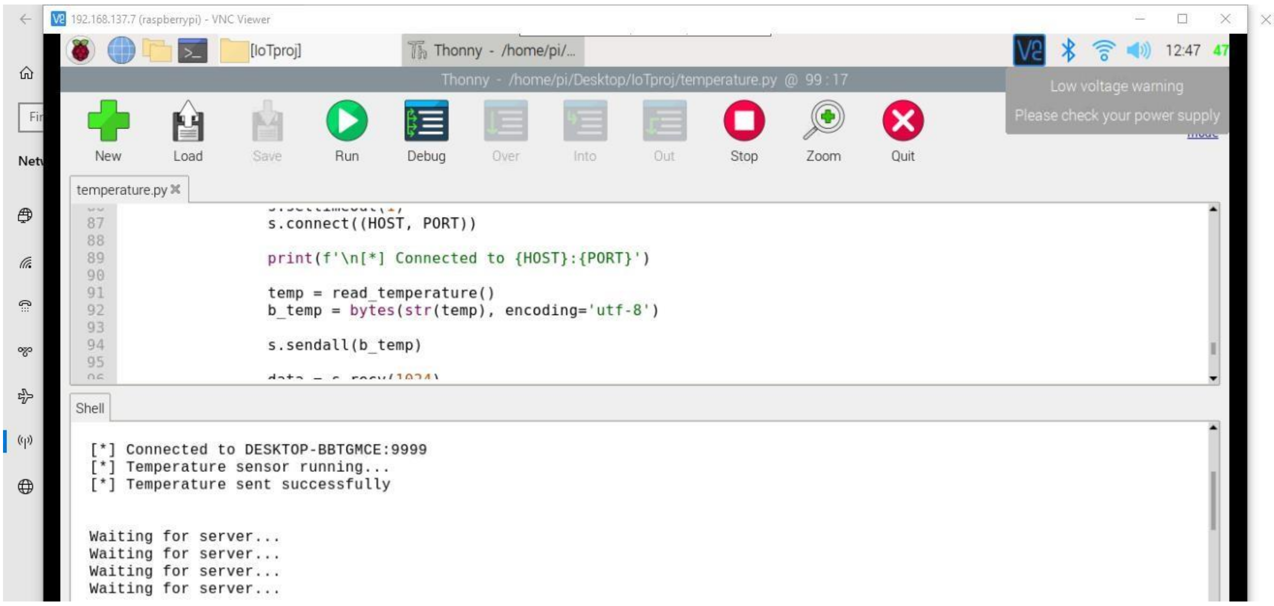

Figure 4-2-0-3: Temperature sent successfully to the desktop application

Postman API Testing

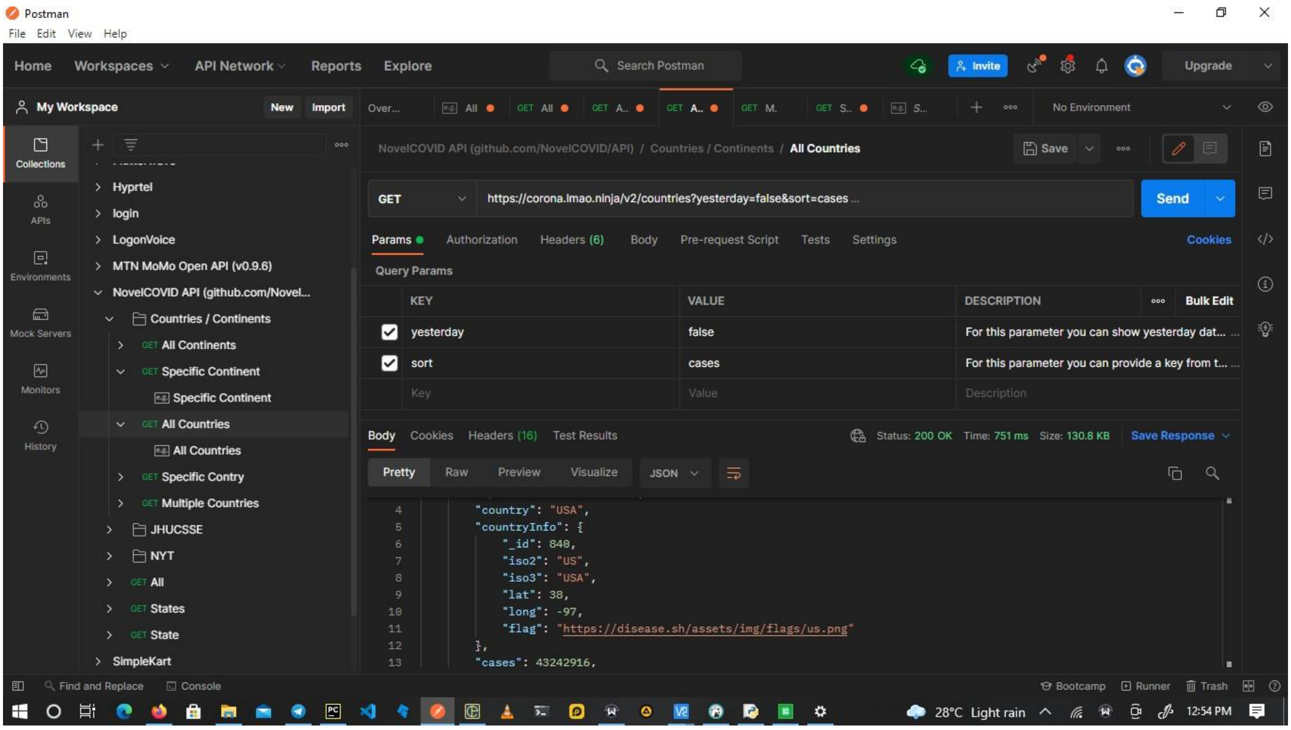

Figure 4-2-0-4: Testing of API endpoing using Postman

4.2.2 Software Subsystem

4.2.2.1 Testing Mask Detection Subsystem

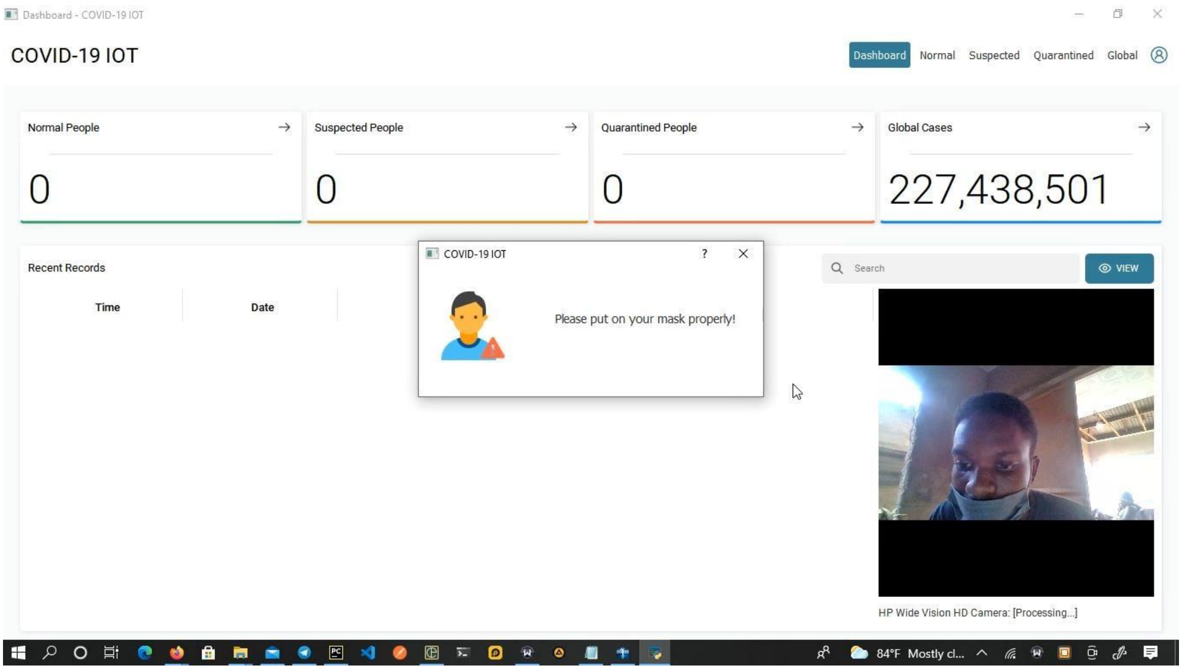

Figure 4-0-5: Result when mask is not worn properly

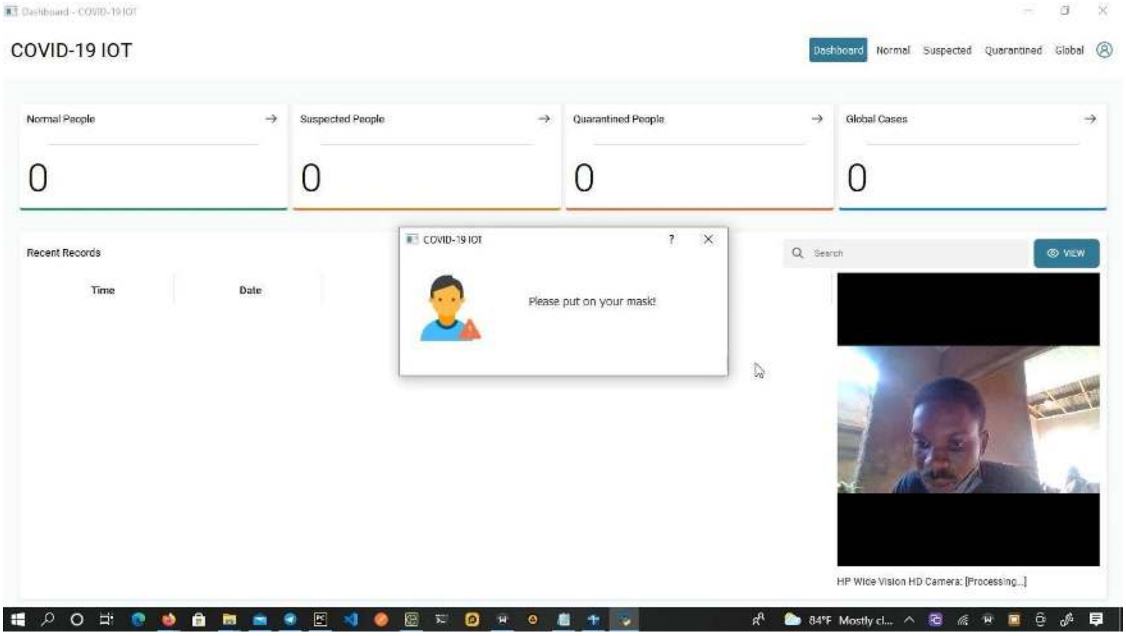

Figure 4-2-5: Results when there is no mask on

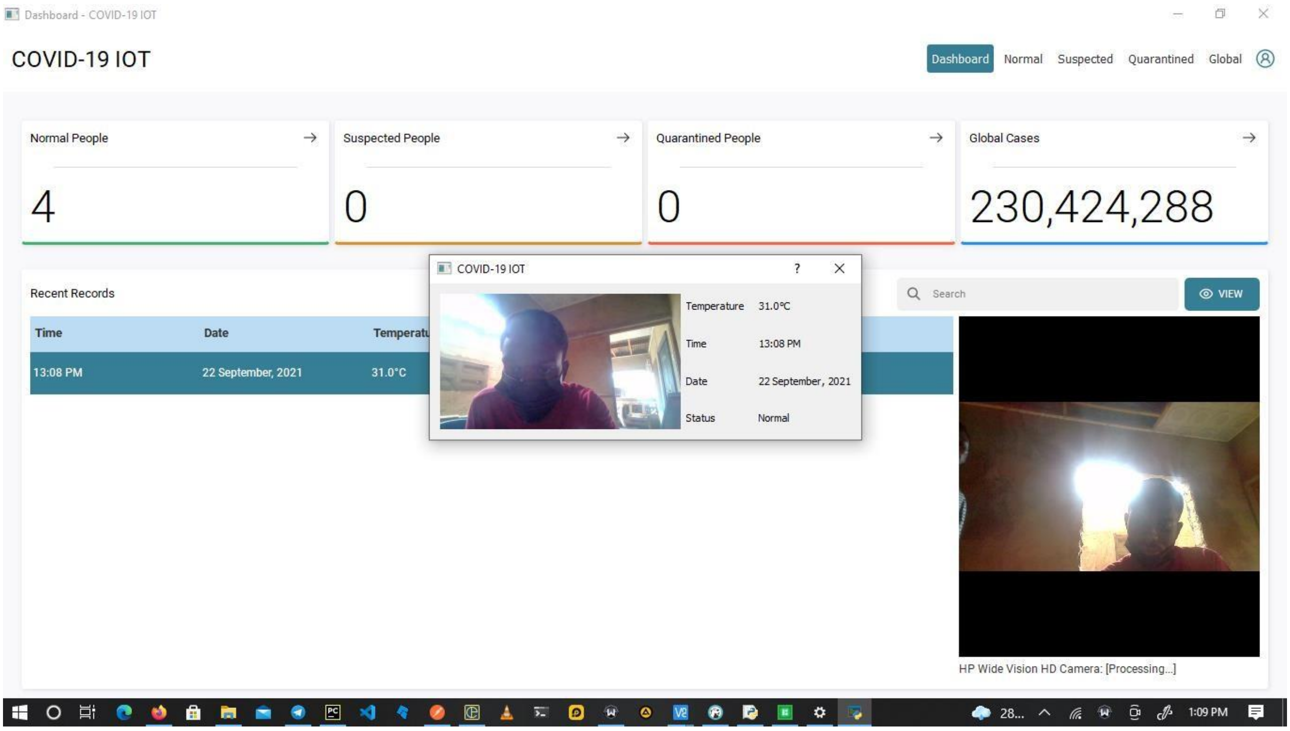

Figure 4-2-6: Result when mask is on

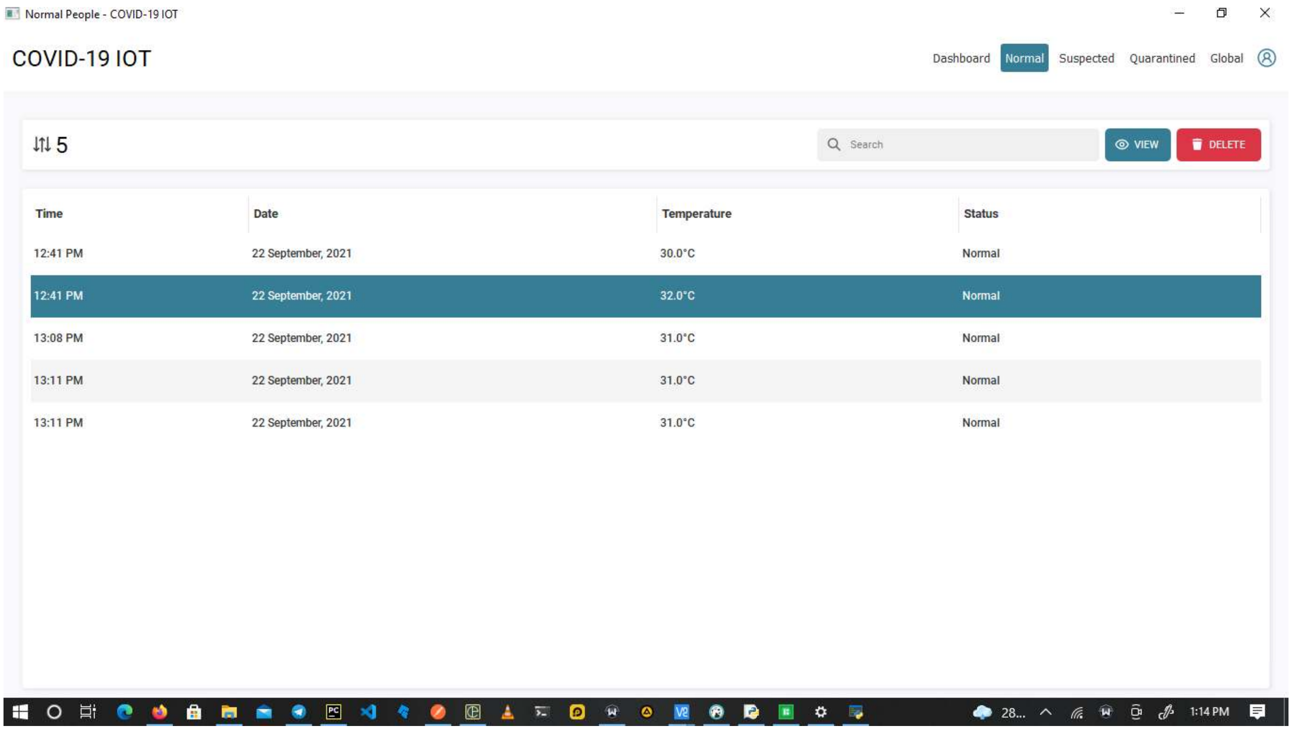

Figure 4-2-7: Table showing normal people with their temperature recordings

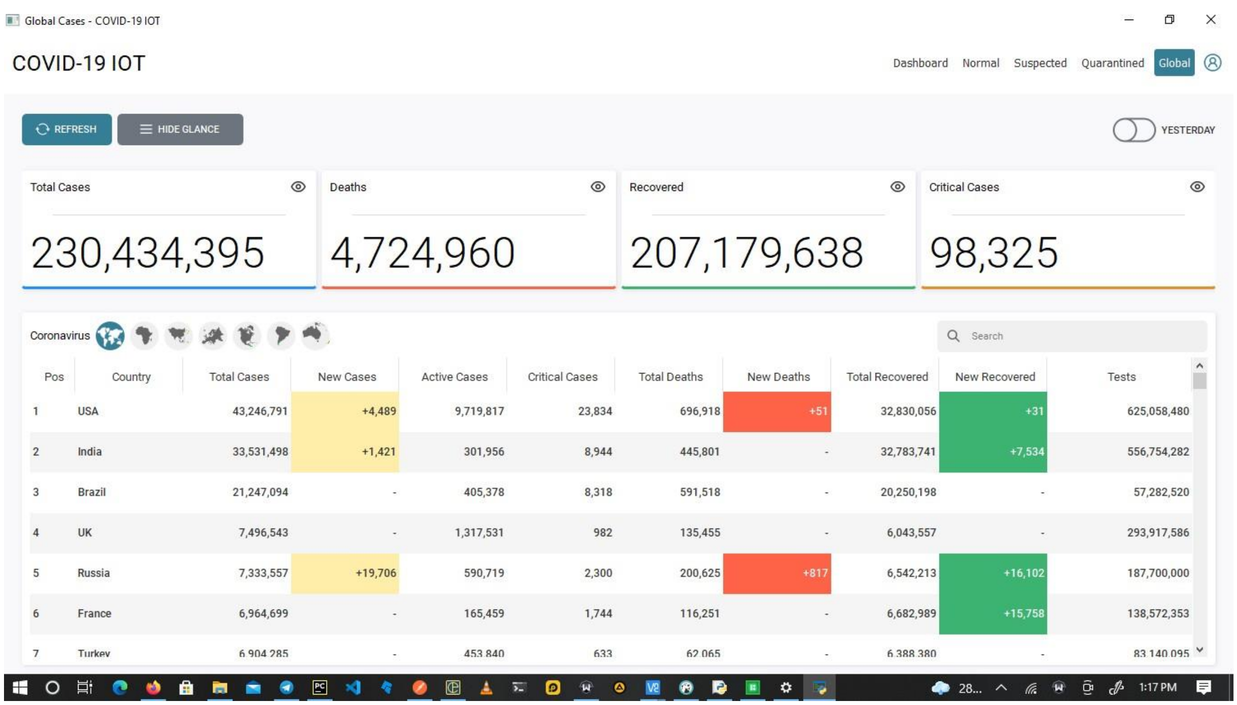

Figure 4-2-8: Global coronavirus update showing the current update

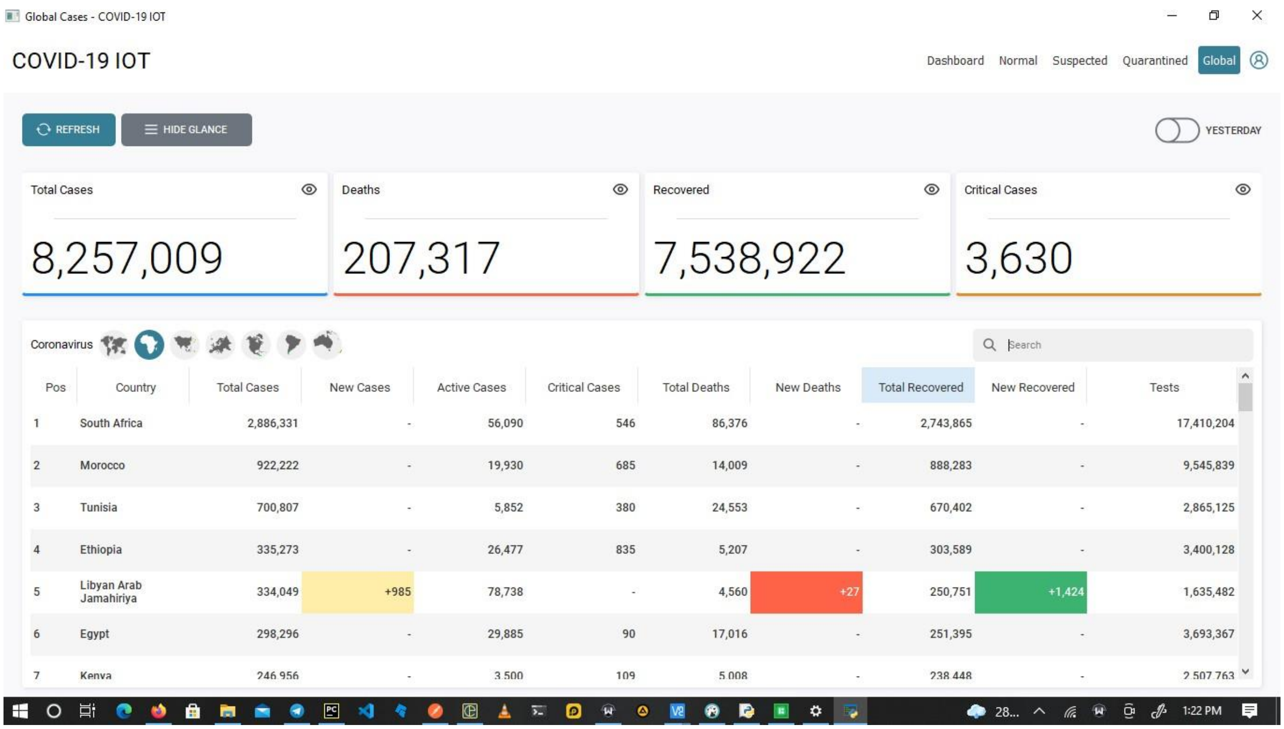

Figure 4-2-9 Real Time Covid updates from Africa

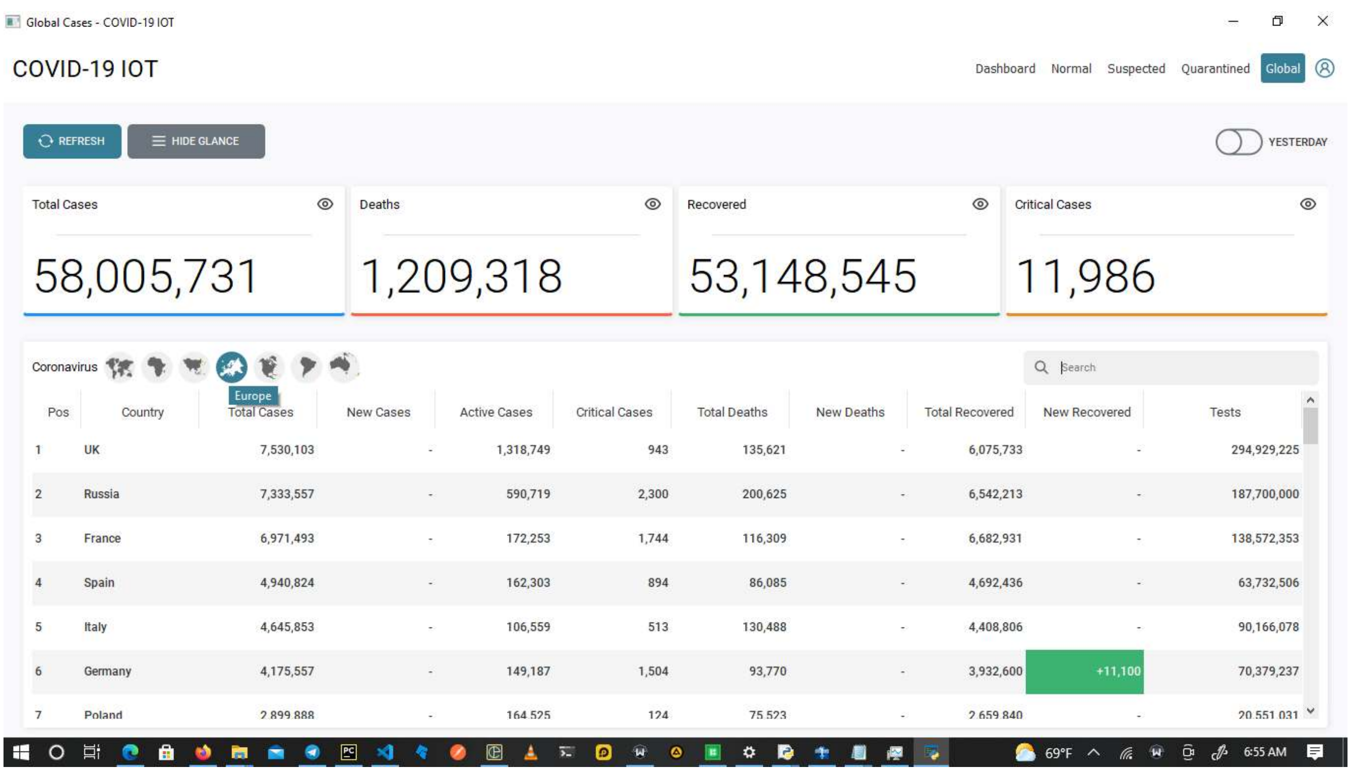

Figure 4-4-10 Real Time Covid updates from Europe

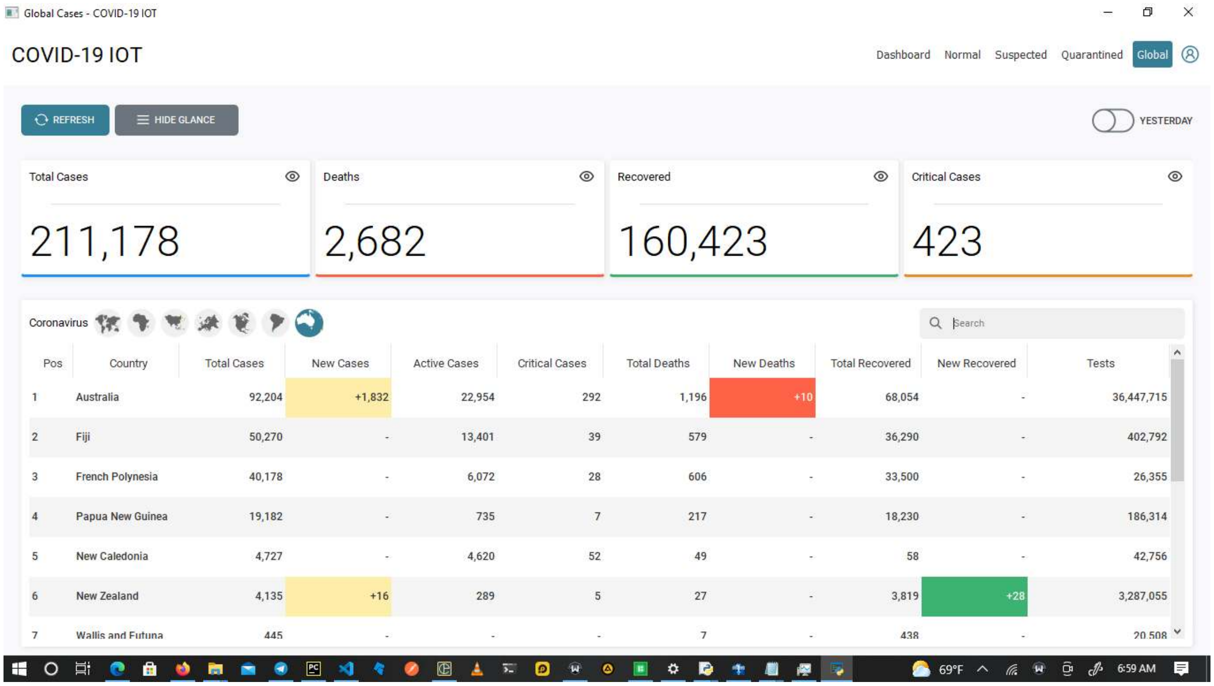

Figure 4-2-11 Real Time Covid Updates from Oceania

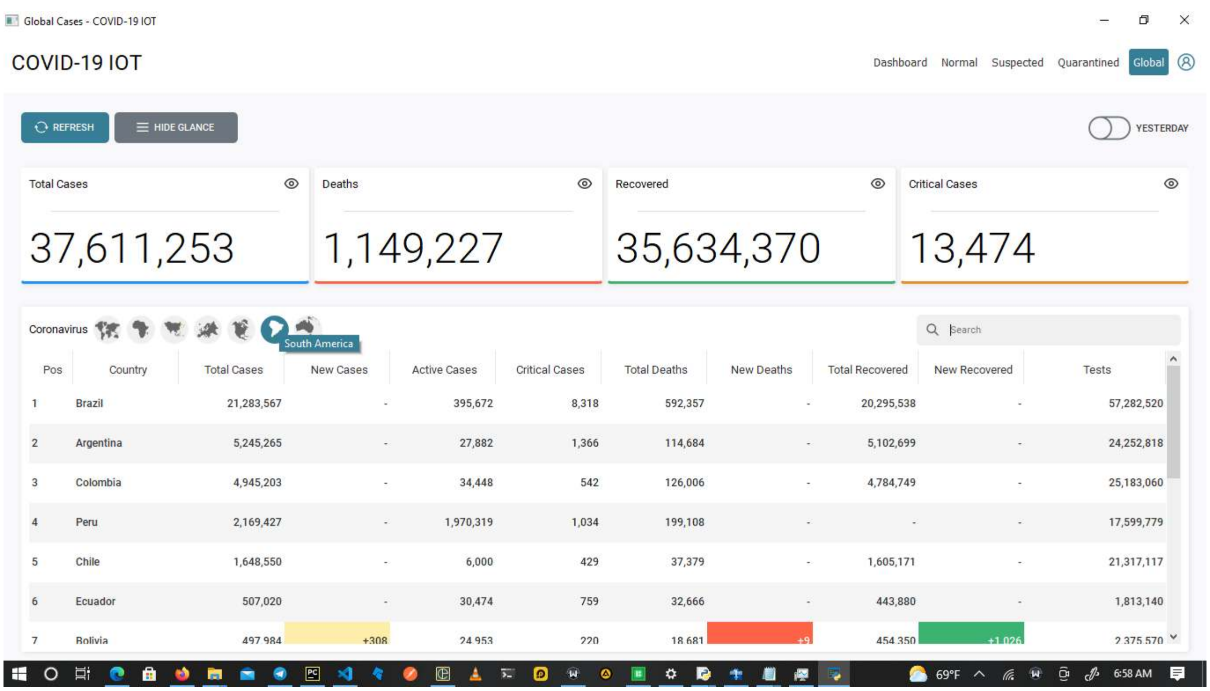

Figure 4-2-12 Real Time Covid Updates from South America

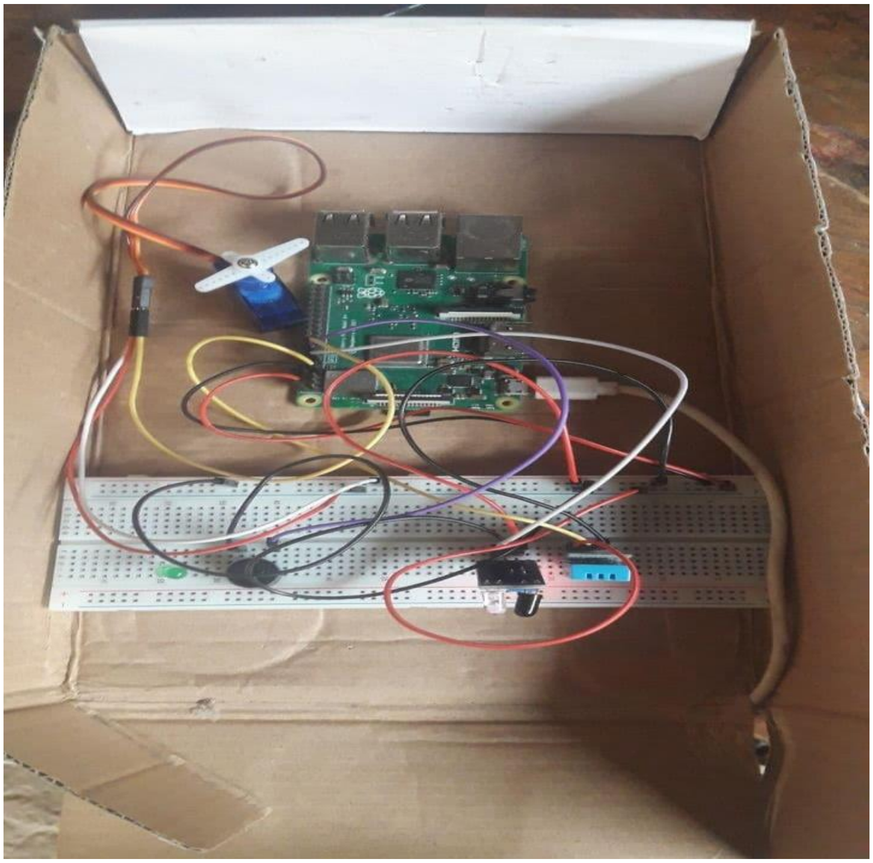

Figure 4-2-13Hardware Setup

4.3 Discussion of results and analysis

Figure 4.1 shows how the connection between the temperature measurement module and the server was tested to see if it works. Here, python sockets were used to establish a connection between the temperature measurement module and the server. A connection was established only when the desktop application sent a request to the server for the temperature of the person to be measured. Figure 4.2 shows the results of a temperature scan being requested by system. Figure 4.3 shows the results obtained when the temperature reading was successfully recorded and sent to the desktop application server. Figure 4.4 shows how Postman software was used to test the API endpoints. Figures 4.5 to 4.7 shows the results of the face mask detection system when a person is in front of the camera. Here, the software uses our trained face mask detection module to detect if faces in the live camera frame have on a mask or not, or do not have on their mask properly and sends the appropriate response to the visitor. The temperature of a person is taken if and only if he has on a mask. Figures 4.8- 4.9 shows the details of a person who has been successfully screened by the system and how the system categorizes people into the normal, suspected and quarantined category.

Figures 4.10 -4.11 shows the results obtained when the admin selects the global button in order to see the real time global statistics of the coronavirus. Figure 4.12 is a picture of the hardware setup of the overall system.

4.4 Performance evaluation and limitations of the system

4.4.1 Performance Evaluation

The table below shows how our system operates when compared to other existing systems.

Table 2:Performance Evaluation of Our System against Existing Systems

System	Detect faces with mask	Detect faces without mask	Detect faces with improper mask	Take temperature measurement	Sound alarm upon high temperature detection	View real time information on an application
Our System	YES	YES	YES	YES	YES	YES
Existing system[2]	YES	YES	NO	YES	YES	NO
Existing system [3]	YES	YES	NO	NO	NO	YES
Existing system[4]	NO	NO	NO	YES	YES	YES
Existing system [5]	YES	YES	NO	NO	NO	YES

Existing system [6]	YES	YES	YES	NO	NO	YES
Existing system [7]	YES	YES	NO	YES	YES	YES

The table below shows how the hardware subsystem performed when various modules were tested

Table 3: *Performance Metrics of Hardware Subsystem*

Hardware Subsystem	Testing	Expected Results	Observation	Performance Percentage
IR SENSOR	The IR sensor was used to detect the presence of a hand.	Hand placed between 0 to 6cm away from the sensor should be detected	Testing out of 10 times, the module worked 10 times.	In percentage, it worked 100% of the time.
TEMPERATURE MEASUREMENT	The DHT11 sensor was used to measure the temperature of a person at the	The DHT11 temperature sensor should be able to	Testing out of 10 times, the module worked properly 8 times.	In percentage, it worked 80% of the time

	door.	detect temperature ranges between 0 to 50 degree Celsius.		
ALERT SYSTEM	The piezo buzzer was used to sound an alarm when temperature was above the required threshold	An alarm should be sounded anytime a temperature above 37 degrees Celsius is measured	Testing out of ten times, the system worked perfectly at all times	Percentage wise, the module worked 100% of the time
Final Hardware System	The overall hardware system was implemented using a Raspberry pi microprocessor interfaced with the various subsystems	Respond to temperature reading requests from the desktop application and send readings to the application in real time	Out of ten times, our system worked 8 times	In percentage, system worked 80% of the time

The table below shows how the software subsystem performed when various modules were tested

Table 4:Performance Metrics of Software system

Software Subsystem	Testing	Expected Results	Observation	Performance Percentage
Face Mask detection	The face mask detection subsystem was tested using a web camera to take images of people at the entrance and determining if they have on a mask	The system should correctly be able to determine if a person has on a mask or not, or has it on improperly	The system worked correctly 9 out of 10 times	Percentage wise, the subsystem worked 90% of the time
Data consumption	The system was tested on how fast it consumes data	The system is to consume less data since it only requires data to load API's to the dashboard	Data consumption was low at all times of testing the system	It worked 100% of the time

Response time	The system was tested to find how fast it responded to real time changes	The system should give real time changes to the user	Out of 10 tests, the response time for the system was appreciably good due to the fastness of the PyQt5 framework	System worked 85% of the time

4.4.2 Limitations of the system

The limitations to this system however is that it does not work well in dark areas and it requires active internet connection for operation.

Chapter 5 – Conclusion and Recommendations

5.0 Introduction

The coronavirus disease has turned into a pandemic and it is presently spreading quickly through direct and indirect contacts among individuals. Manual systems of ensuring Covid 19 protocols are observed which are being used in our homes and public places might turn into a wellspring of the spread of infection of Covid 19. Presently this infection will remain in our lives, and we need to live with it, however need to take safety measures stringently to break the chain of this infection. The rationale of our work comes from individuals defying the principles that are obligatory to stop the spread of Covid. Our system focuses on face mask detection architecture, temperature monitoring architecture and an app for contact tracing.

5.1 Conclusions

In our work, an IoT enabled smart door is built up to recognize facemasks and to check body temperature that can upgrade public security. Also our work has a web app that is used for contact tracing and also provides update of global statistics of the Covid 19. This will assist with decreasing labor while likewise giving an additional layer of protection against the spread of Covid 19 disease. This model uses desktop application to detect face mask and raspberry pi to monitor body temperature.

5.2 Recommendations

We recommend that our future expansion will include social distancing between individuals. we plan on achieving that by using a pre trained deep learning algorithms. Also we will focus on implementing an automatic hand sanitizer Dispenser as regularly sanitizing of hands is one of the main guidelines prescribed by WHO. we plan to achieve this using an Arduino, ultrasonic sensor, water pump and hand sanitizer

5.3 Observations

With this project, individuals will be able to observe the covid 19 protocols by wearing their nose mask properly and also having their body temperature checked. Also the web app admin should always be connected to the internet so as to check the status of any individual that passes through our door.

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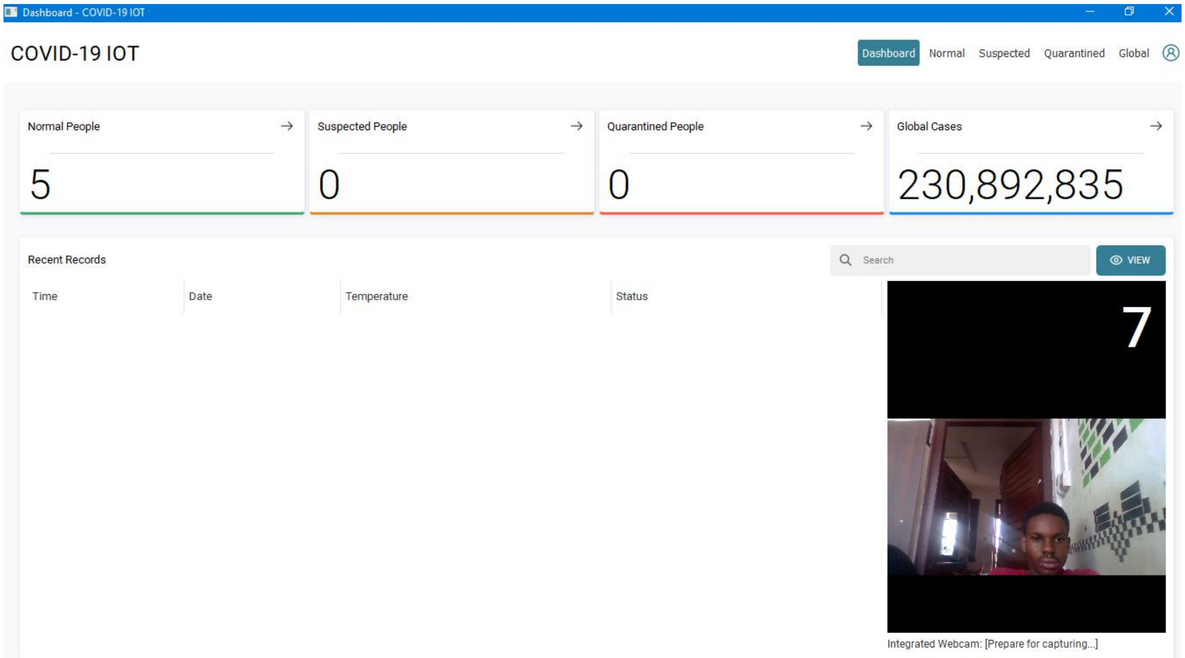
APPENDICES

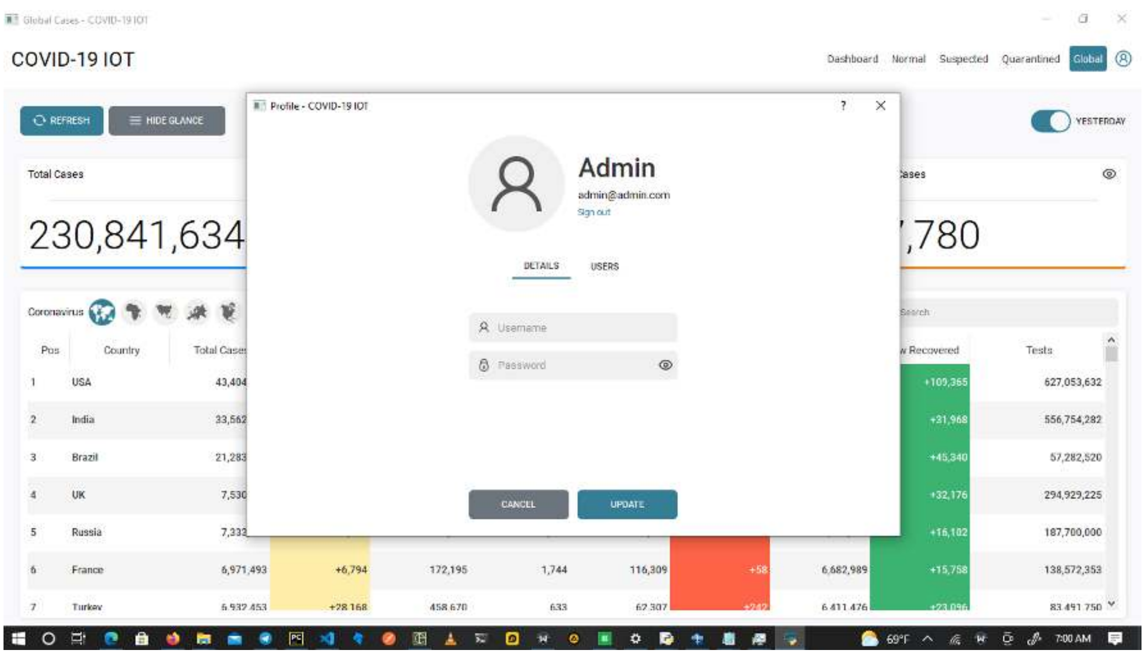
APPENDIX A- INTERFACES FOR DESKTOP APPLICATION

- **Login Page**

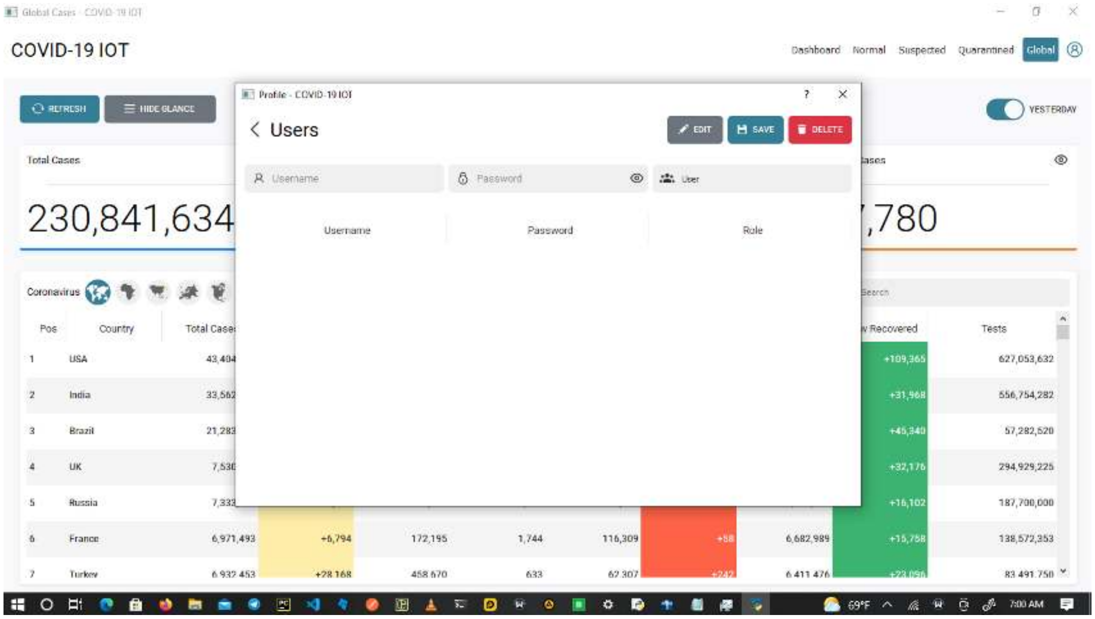
- **Dashboard**



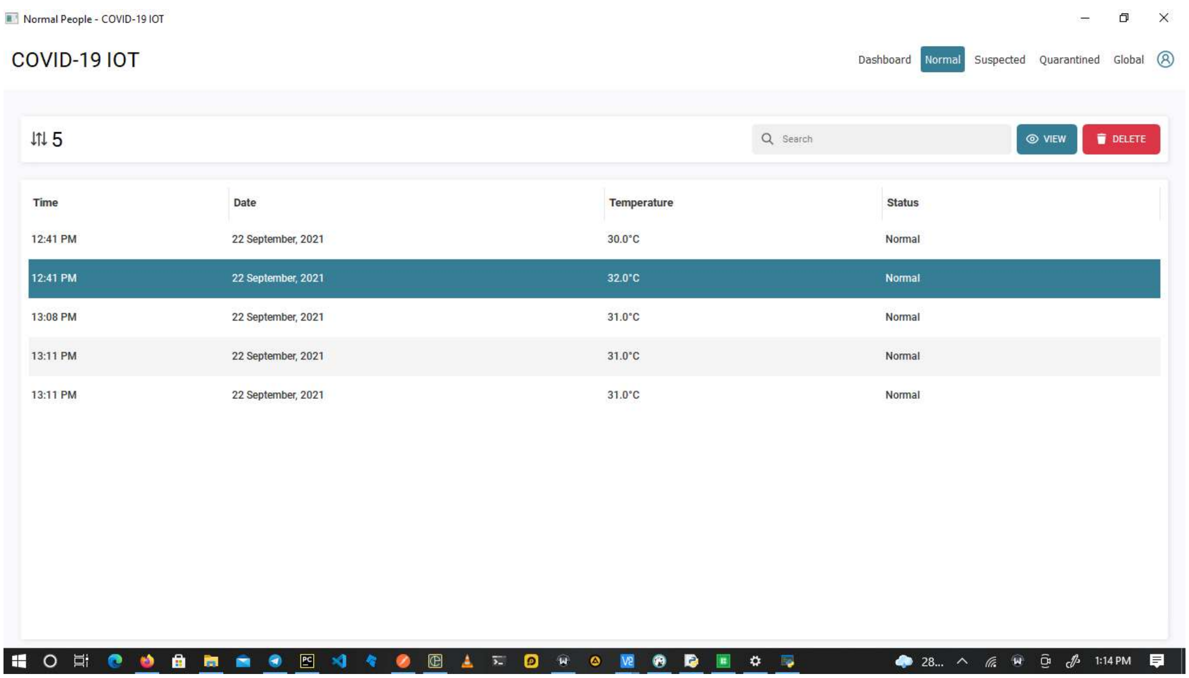
- **User Profile**



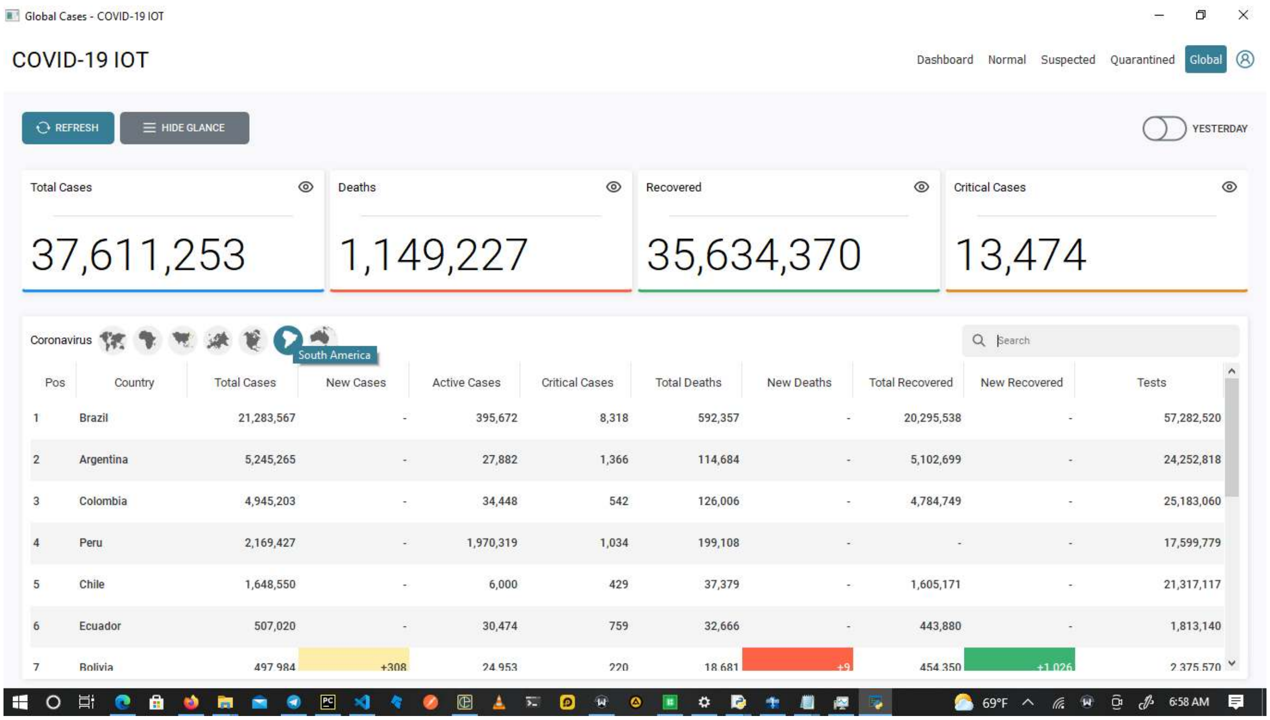
- **User’s table**



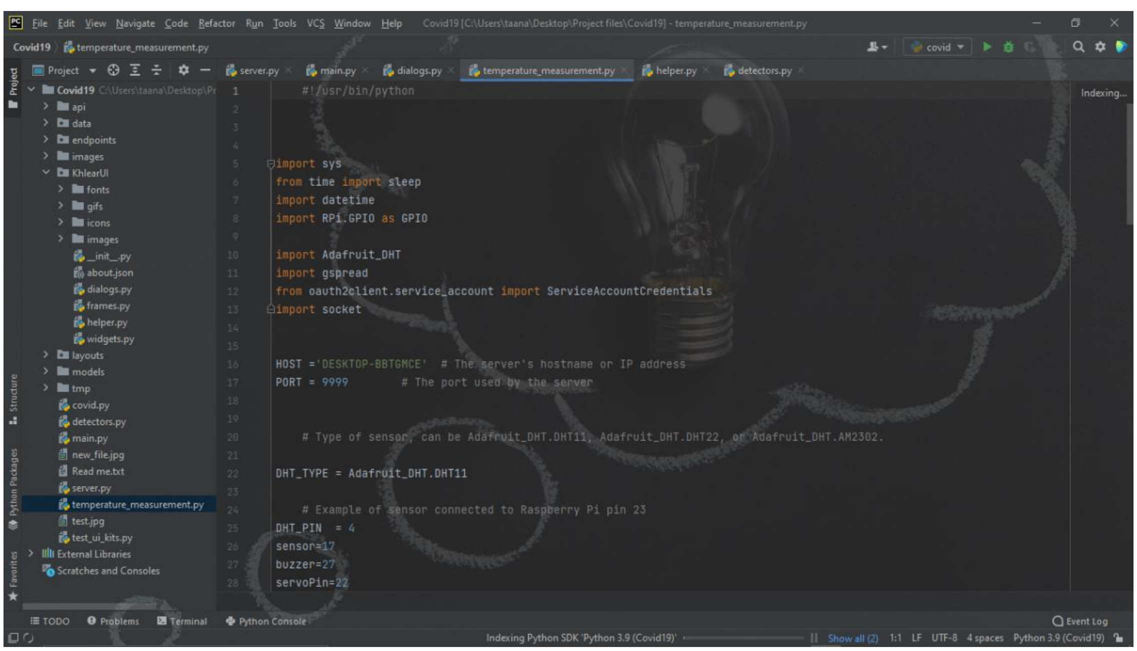
- **Normal people data**

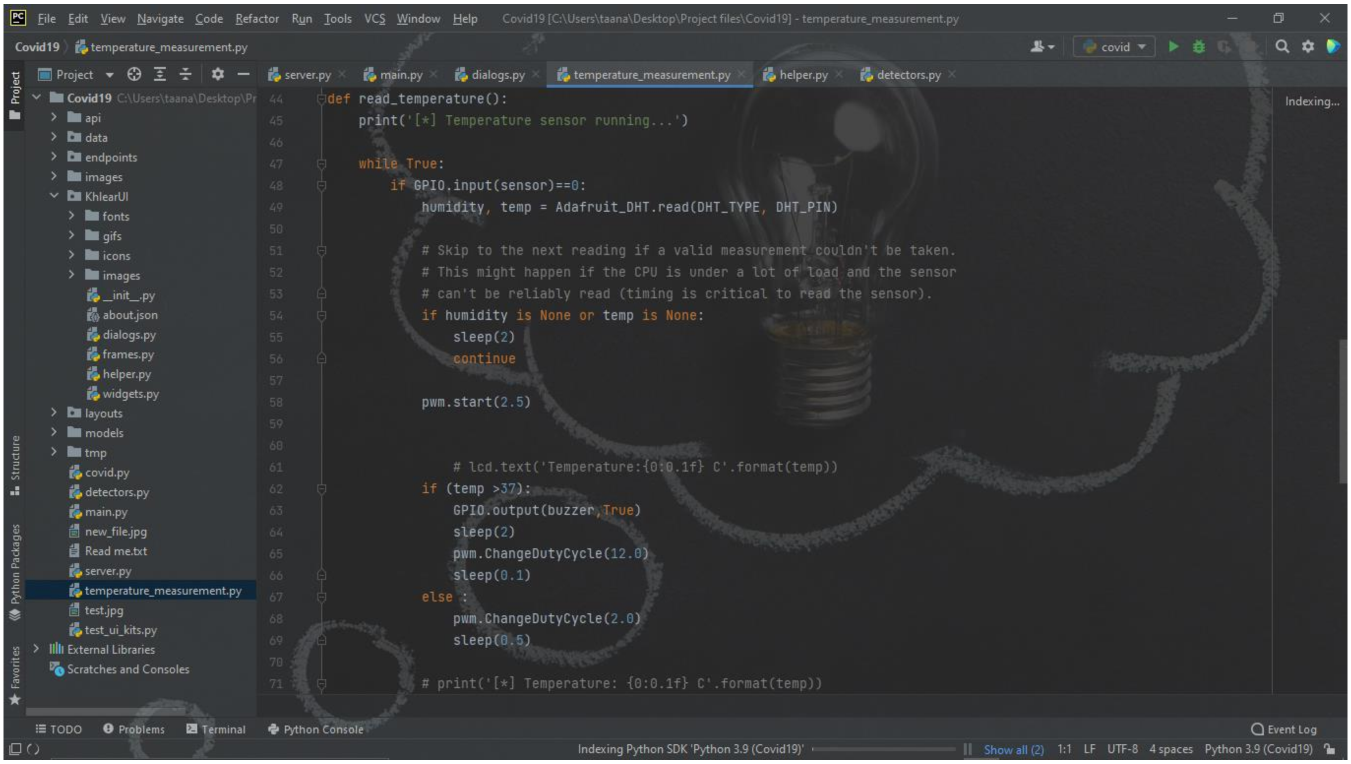


- **GLOBAL CORONA VIRUS UPDATE DASHBOARD**

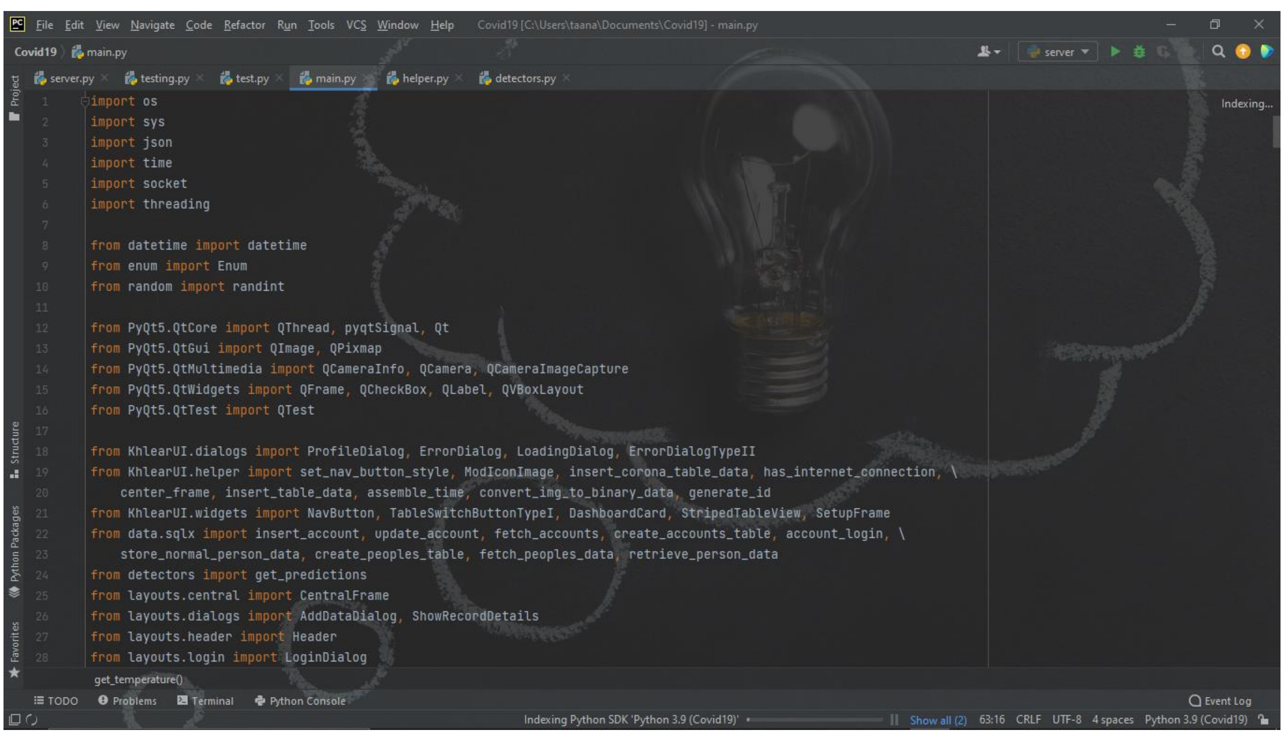


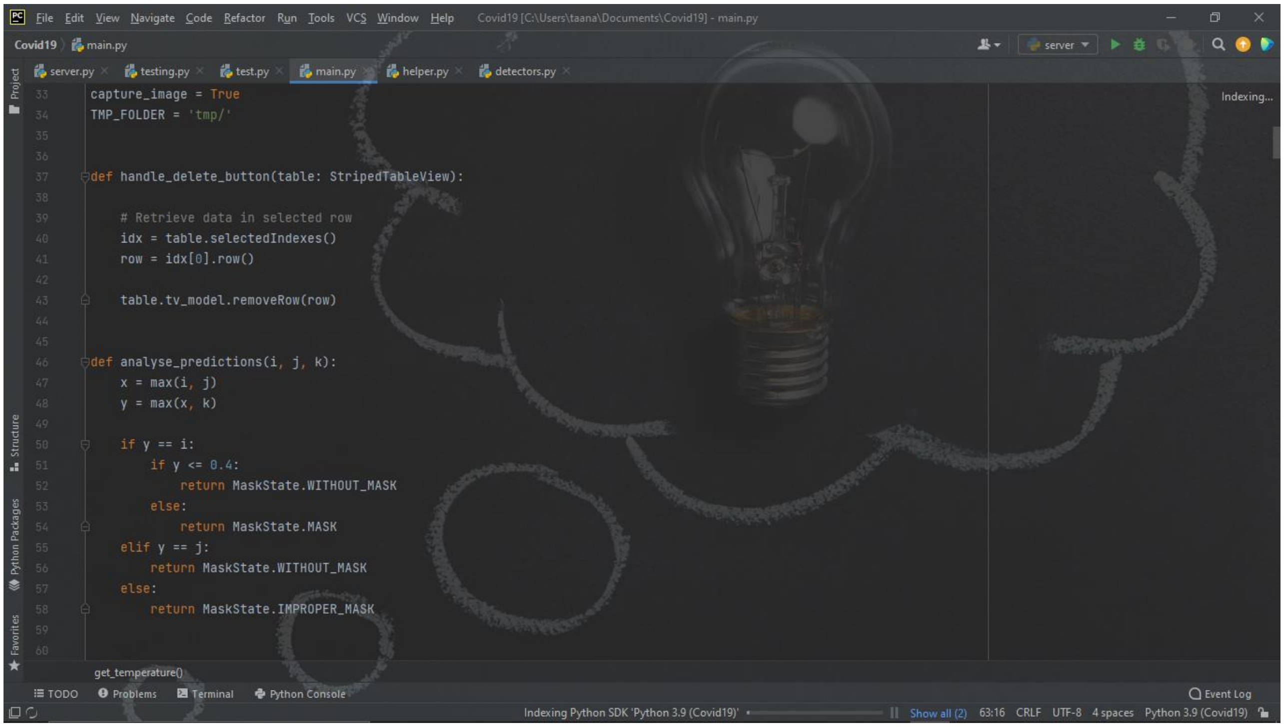
APPENDIX B – RASPBERRY PI CODE SNIPPETS FOR HARDWARE MODULE



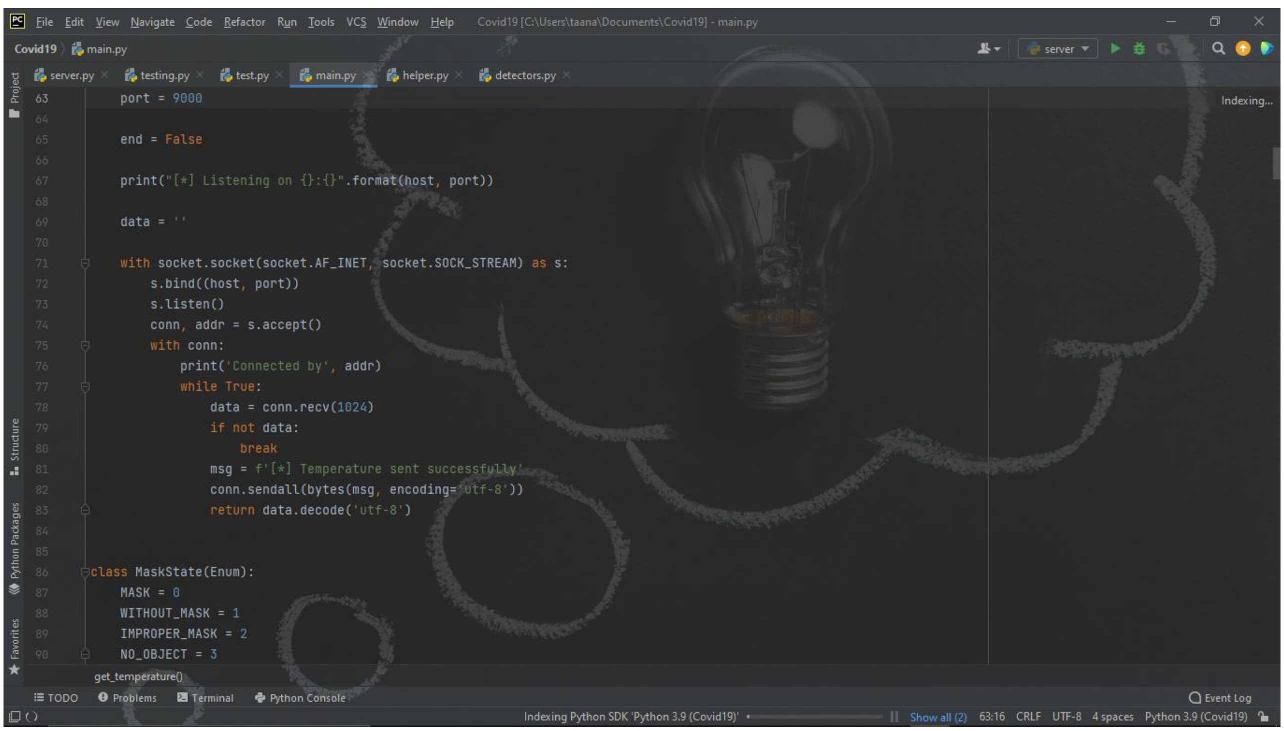


APPENDIX C- CODE SNIPPETS FOR DESKTOP APPLICATION

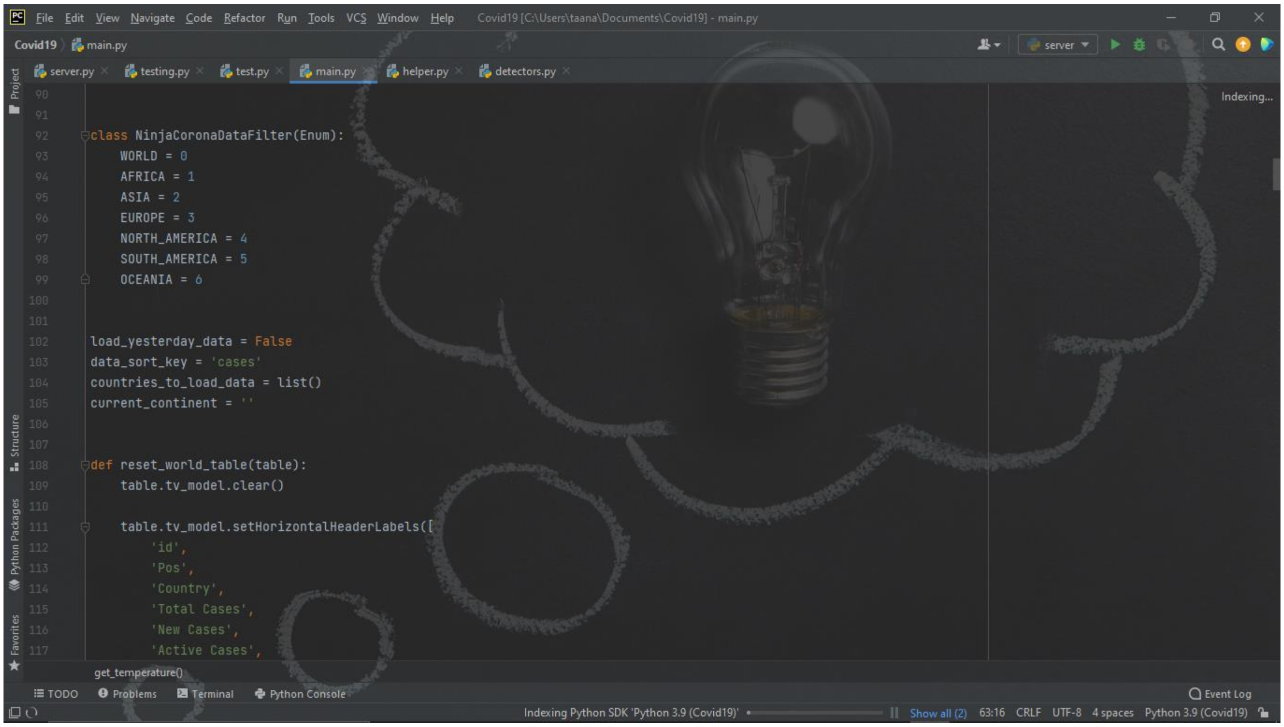




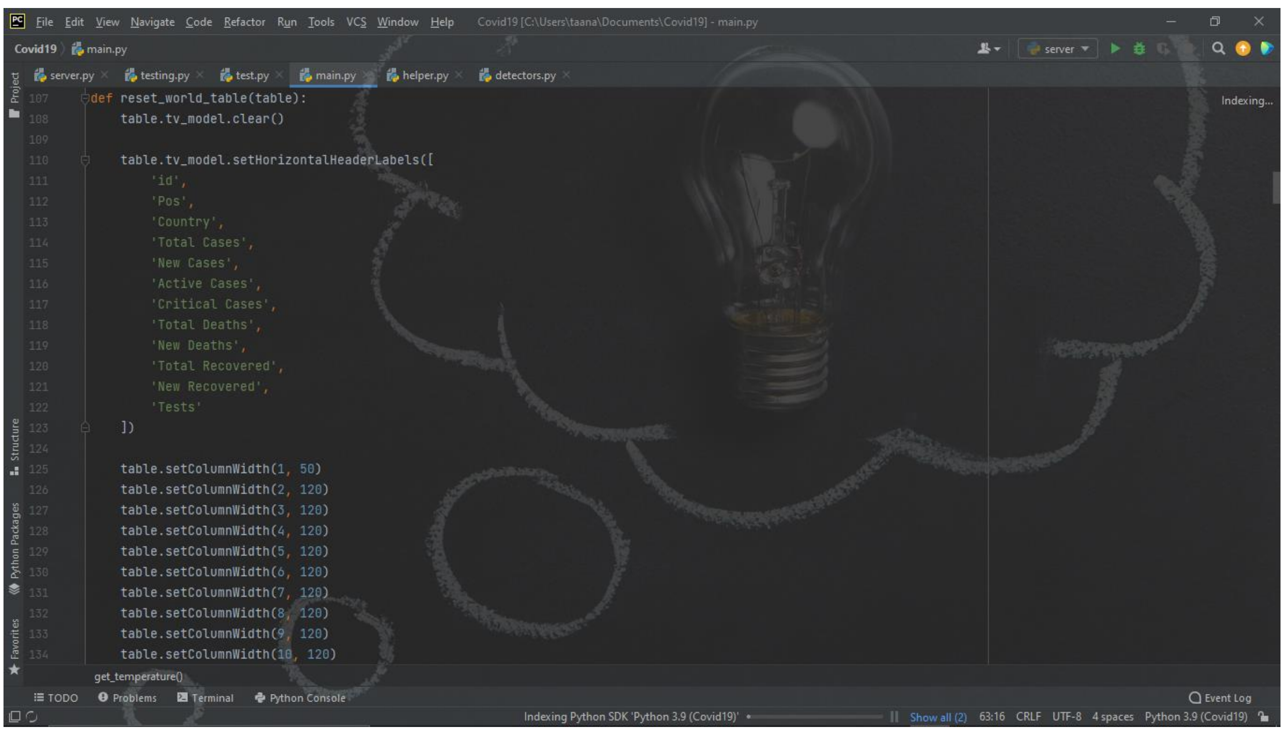
capture_image = True
TMP_FOLDER = 'tmp/'
def handle_delete_button(table: StripedTableView):
# Retrieve data in selected row
idx = table.selectedIndexes()
row = idx[0].row()
table.tv_model.removeRow(row)
def analyse_predictions(i, j, k):
x = max(i, j)
y = max(x, k)
if y == i:
if y <= 0.4:
return MaskState.WITHOUT_MASK
else:
return MaskState.MASK
elif y == j:
return MaskState.WITHOUT_MASK
else:
return MaskState.IMPROPER_MASK



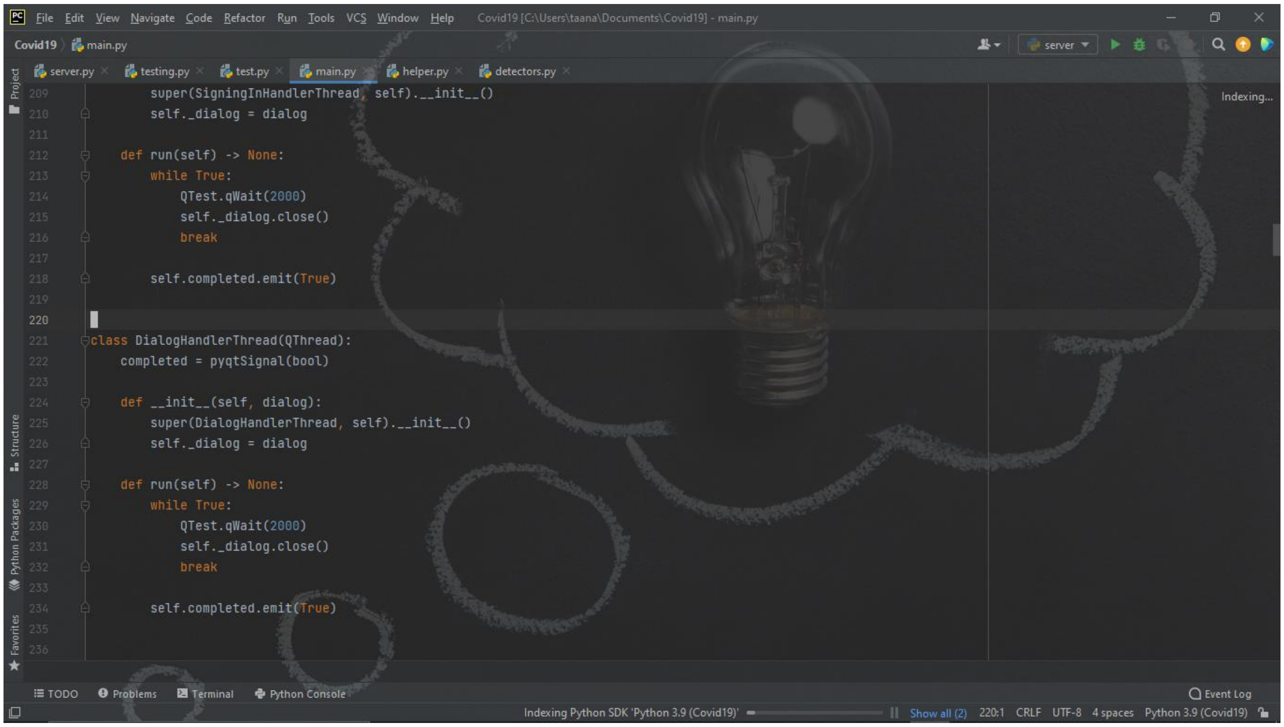
port = 9000
end = False
print("[*] Listening on {}:{}".format(host, port))
data = ''
with socket.socket(socket.AF_INET, socket.SOCK_STREAM) as s:
s.bind((host, port))
s.listen()
conn, addr = s.accept()
with conn:
print('Connected by', addr)
while True:
data = conn.recv(1024)
if not data:
break
msg = f'[*] Temperature sent successfully'
conn.sendall(bytes(msg, encoding='utf-8'))
return data.decode('utf-8')
class MaskState(Enum):
MASK = 0
WITHOUT_MASK = 1
IMPROPER_MASK = 2
NO_OBJECT = 3

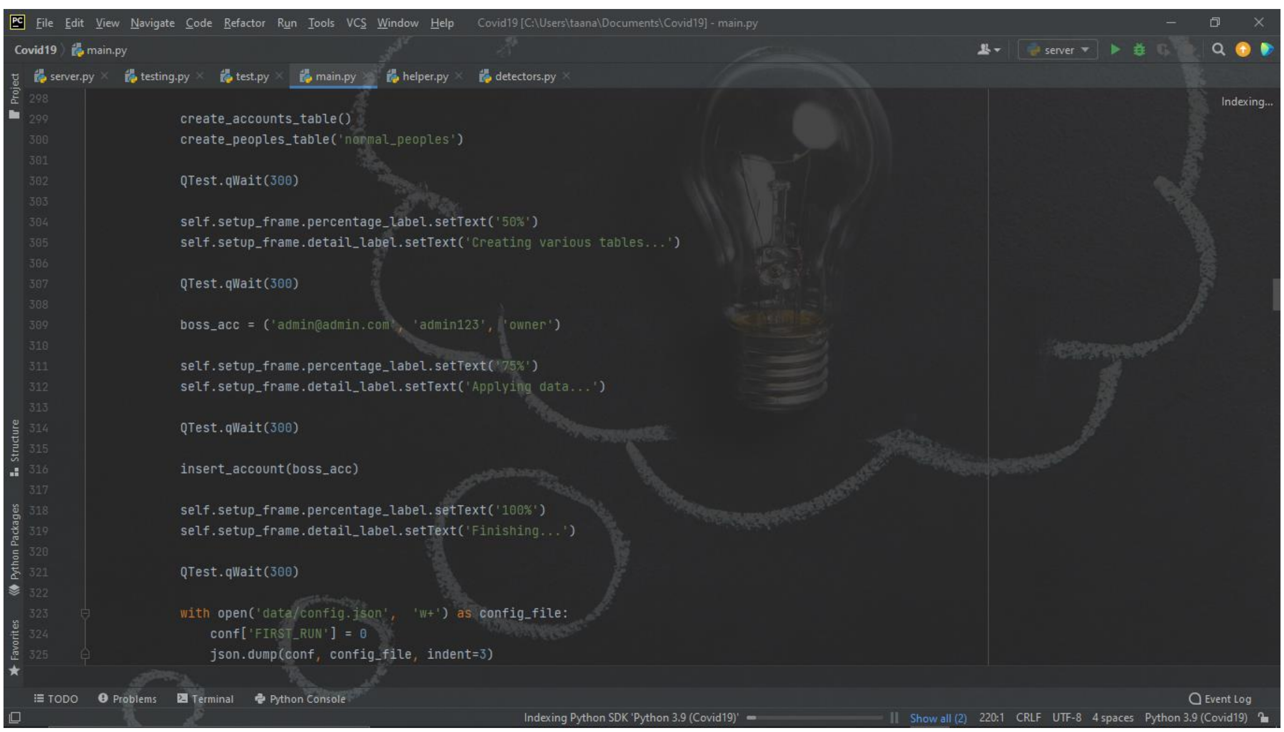


```python


class NinjaCoronaDataFilter(Enum):
    WORLD = 0
    AFRICA = 1
    ASIA = 2
    EUROPE = 3
    NORTH_AMERICA = 4
    SOUTH_AMERICA = 5
    OCEANIA = 6


load_yesterday_data = False
data_sort_key = 'cases'
countries_to_load_data = list()
current_continent = ''


def reset_world_table(table):
    table.tv_model.clear()

    table.tv_model.setHorizontalHeaderLabels([
        'id',
        'Pos',
        'Country',
        'Total Cases',
        'New Cases',
        'Active Cases',
```



```python
def reset_world_table(table):
    table.tv_model.clear()

    table.tv_model.setHorizontalHeaderLabels([
        'id',
        'Pos',
        'Country',
        'Total Cases',
        'New Cases',
        'Active Cases',
        'Critical Cases',
        'Total Deaths',
        'New Deaths',
        'Total Recovered',
        'New Recovered',
        'Tests'
    ])

    table.setColumnWidth(1, 50)
    table.setColumnWidth(2, 120)
    table.setColumnWidth(3, 120)
    table.setColumnWidth(4, 120)
    table.setColumnWidth(5, 120)
    table.setColumnWidth(6, 120)
    table.setColumnWidth(7, 120)
    table.setColumnWidth(8, 120)
    table.setColumnWidth(9, 120)
    table.setColumnWidth(10, 120)
```


super(SigningInHandlerThread, self).__init__()
self._dialog = dialog
def run(self) -> None:
while True:
QTest.qWait(2000)
self._dialog.close()
break
self.completed.emit(True)
class DialogHandlerThread(QThread):
completed = pyqtSignal(bool)
def __init__(self, dialog):
super(DialogHandlerThread, self).__init__()
self._dialog = dialog
def run(self) -> None:
while True:
QTest.qWait(2000)
self._dialog.close()
break
self.completed.emit(True)


create_accounts_table()
create_peoples_table('normal_peoples')
QTest.qWait(300)
self.setup_frame.percentage_label.setText('50%')
self.setup_frame.detail_label.setText('Creating various tables...')
QTest.qWait(300)
boss_acc = ('admin@admin.com', 'admin123', 'owner')
self.setup_frame.percentage_label.setText('75%')
self.setup_frame.detail_label.setText('Applying data...')
QTest.qWait(300)
insert_account(boss_acc)
self.setup_frame.percentage_label.setText('100%')
self.setup_frame.detail_label.setText('Finishing...')
QTest.qWait(300)
with open('data/config.json', 'w+') as config_file:
conf['FIRST_RUN'] = 0
json.dump(conf, config_file, indent=3)